Informational Power Through Pivotality: When a Hegemon May Choose Consensus

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Abstract

When does a powerful state prefer consensus to majority rule in an international organization? I develop a formal model of institutional design in which a hegemon has private information about the value of multilateral cooperation. Under majority rule, weaker states can form winning coalitions without the hegemon, so its private information has limited strategic value. Under unanimity, every agreement requires the hegemon's approval, forcing weaker states to decide how much to offer without knowing how valuable cooperation really is. This uncertainty works in the hegemon's favor: weaker states pay more than necessary to secure agreement, and majority rule eliminates this advantage entirely. In calibrated examples, unanimity gives the hegemon between 27% and 41% higher payoffs than majority rule near the screening cutoff, with the advantage declining as uncertainty decreases. This screening advantage means that, once cooperation is promising enough for weaker states to participate under unanimity, the hegemon strictly prefers it. Below that threshold, majority can dominate—but only because it makes participation easier, not because it gives better terms. Consensus, in this account, is not a concession to weaker states but an institutional arrangement that makes informational advantage politically productive—offering a distributive explanation for consensus rules in organizations such as the WTO.

1 Introduction

Most international organizations operate by consensus, even when powerful states drive their creation (Gould 2016). The international arena is where material asymmetries should matter most relative to domestic politics. If institutional rules reflect power, powerful states might be expected to favor arrangements that let them turn that power into a distributive advantage. Why would a powerful state ever prefer consensus over majority rule, where it could use agenda power to impose its preference without any state having veto power?

Under majority rule, a powerful actor can form coalitions, exclude rivals, and keep the surplus (Baron and Ferejohn 1989; Kalandrakis 2006; Eraslan and Evdokimov 2019). Consensus does the opposite: by giving every participant a veto, it forces powerful states to surrender the tools they might value most. Yet international organizations are built around consensus, even when powerful states drive their creation (Gould 2016; Steinberg 2002; Koremenos, Lipson, and Snidal 2001). Existing accounts usually see consensus either as constraining powerful states through institutional self-binding (Keohane 1984; Ikenberry 2001) or as a façade for informal power that operates irrespective of the formal rule (Steinberg 2002; Stone 2011; Gruber 2000). Neither explains when and why a hegemon would prefer unanimity to a rule that allows exclusion.

This paper theorizes when and why a hegemon may prefer unanimity to majority rule in international organizations. The central claim is that consensus is an institutional technology of power. I show that consensus can make informational power more politically productive than the coalition arithmetic of majority rule. By informational power I mean the bargaining advantage that a privately informed actor derives from being pivotal: when other players must secure its approval without knowing its type, uncertainty itself becomes a source of rent.

To formalize the argument, I develop a model in which a hegemon chooses between majority and unanimity. Proposal rights are symmetric under both rules, isolating the effect of the voting rule itself. The hegemon privately observes the value of cooperation before weaker states decide whether to enter. The central difference is that the two rules reward this informational advantage differently. Under majority rule, weak states can form coalitions without making the hegemon pivotal, eliminating the screening problem. Under unanimity, weak states must include the hegemon in every proposal and decide how much to offer without knowing the probability of acceptance. This screening generates a cutoff at which weak proposers switch from aggressive to conservative treatment of the hegemon, giving the hegemon a strictly higher conditional payoff than under majority at every level of uncertainty. The institutional comparison reduces to a single distinction: unanimity dominates wherever it can sustain entry, and majority's only advantage is wider institutional viability.

The substantive implication is that consensus is most valuable to a hegemon when the prospects of multilateral cooperation are promising enough that weaker states are willing to come to the table. Under those conditions, unanimity amplifies the hegemon's informational advantage. On the other hand, when cooperation looks too uncertain for weaker states to participate at all, majority rule can be preferred because it lowers the bar for participation.

2 Consensus, Information, and Institutional Design

The paper contributes to the literature on institutional design in international organizations. The dominant explanations for consensus can be grouped into three categories. Consensus as self-binding: powerful states accept formal constraints to make credible commitments, encouraging participation by weaker partners (Keohane 1984; Ikenberry 2001). Consensus as irrelevant: informal agenda power, control over drafting, and asymmetric resources allow powerful states to shape outcomes regardless of the formal voting rule (Steinberg 2002; Stone 2011; Gruber 2000). Consensus as rational design: voting rules are chosen to match the cooperation problem an organization faces (Koremenos, Lipson, and Snidal 2001). Gould (2016) provides the most comprehensive empirical account, documenting the prevalence and variation of consensus across international organizations. This paper offers a distinct account: consensus as a technology that activates informational power. The mechanism does not require informal influence, does not serve as self-binding, and is not chosen to solve commitment problems. It is preferred because it makes the hegemon's private information more valuable.

The model builds on the Baron-Ferejohn framework of legislative bargaining (Baron and Ferejohn 1989). Kalandrakis (2006) shows that proposal rights are the primary determinant of political power in such models; the present paper holds proposal rights symmetric to isolate the effect of the voting rule. Eraslan and Evdokimov (2019) survey the broader literature. The key departure is the combination of asymmetric information about the value of cooperation with a choice between voting rules, a combination that generates endogenous screening under unanimity but not under majority.

The paper is most closely related to formal work on information in voting and bargaining. Bardhi and Guo (2018) study information design under unanimity, showing how a persuader can manipulate the beliefs of multiple receivers who must unanimously agree. Their framework does not include bargaining over a divisible surplus, endogenous screening by the uninformed side, or a comparison between voting rules. Kim, Kim, and Van Weelden (2025) study persuasion in veto bargaining, analyzing how an information designer augments agenda-setting power when a proposer faces a veto player. Their model is bilateral and takes the veto rule as given. Neither paper addresses the question that is central here: how the voting rule itself determines whether screening occurs. In both, information design is the instrument; in the present paper, the voting rule is the instrument, and screening is the endogenous consequence of pivotality under uncertainty.

The informational mechanism generates a prediction that separates it from the main alternatives. Because screening rents require that weaker states face genuine uncertainty about the value of cooperation, the model predicts a negative cross-sectional association between consensus rules and transparent distributive stakes. Financial institutions, where capital contributions and loan terms make the pie calculable, should not adopt consensus—and overwhelmingly use weighted majority. Regulatory and standard-setting bodies, where consequences are difficult to forecast, should be the most frequent adopters—and are (Gould 2016). Legitimacy accounts do not condition on informational structure. Self-binding accounts predict consensus where commitment problems are severe, regardless of transparency. Informal power accounts cannot explain why consensus is absent precisely where powerful states have the strongest informal leverage.

3 A Motivating Example

Before turning to the general model, consider a simplified special case with three players and one bargaining round to build intuition for the mechanism. The general model (Section 4) extends this to N players, two rounds of Baron-Ferejohn bargaining, and entry costs. There is one hegemon H and two weak states W_1, W_2. The value of cooperation is either $V(0) = 1$ (low) or $V(1) = 2$ (high); H observes the state, the weak states do not. Each weak state's outside option is 0; H's outside option is $\alpha V(\theta)$ with $\alpha = 0.2$. A proposer is selected uniformly at random; to illustrate the screening mechanism, I focus on the case where a weak state is selected and makes a take-it-or-leave-it offer specifying H's share y_H. If H rejects, each player receives their disagreement payoff.

Throughout the paper, I model consensus as unanimity over a binary accept/reject decision. This abstracts the distinction between unanimity (everyone must explicitly consent to an agreement) and consensus (no one explicitly blocks a deal), but captures the central strategic feature: any member can block agreement.

Under majority rule, W_1 and W_2 can pass a proposal without H's vote. The proposing weak state excludes H from the coalition, and H captures $0.2V(\theta)$ from its bilateral alternative regardless of what W believes. The hegemon's expected payoff is $0.2+0.2\mu$, where $\mu = \Pr(\theta = 1)$. The payoff is linear in beliefs, with no strategic discontinuity. Because the hegemon's vote is never needed, its private information creates no leverage.

Under unanimity, H's vote is required, and the proposing weak state faces a dilemma. It must decide how much to offer H without knowing θ:

- An aggressive offer ($y_H = 0.2$) pays the hegemon its low-state outside option. Type $\theta = 0$ is indifferent and accepts; type $\theta = 1$ rejects, since it wants 0.4. The weak proposer gets 0.8 if $\theta = 0$ and 0 if $\theta = 1$.
- A conservative offer ($y_H = 0.4$) pays the hegemon its high-state outside option. Both types accept. The weak proposer gets 0.6 if $\theta = 0$ and 1.6 if $\theta = 1$.

The weak state prefers the aggressive offer when $0.8(1-\mu) > 0.6(1-\mu)+1.6\mu$, which simplifies to $\mu < 1/9$. This defines a screening cutoff: the belief at which the weak proposer switches from aggressive to conservative treatment of the hegemon.

The switch in behavior creates a discrete jump in the hegemon's expected payoff. Below the cutoff, the hegemon earns $0.2 + 0.2\mu$, the same as under majority. Above it, the expected payoff jumps to 0.4, constant and independent of μ. The source of the jump is overpayment: on the conservative branch, type $\theta = 0$ receives 0.4 but would accept 0.2. Weak states cannot avoid this cost because they must secure the hegemon's vote without knowing the type. At the cutoff $\mu^* = 1/9$, the jump is $8/45 \approx 0.18$, about 16% of expected surplus.

The screening jump gives the hegemon a strictly higher conditional payoff under unanimity than under majority for every belief above the cutoff. The general model (Section 4 onwards) shows this holds across the entire belief space (Theorem~1).

The general model enriches this logic in several respects. With N players and two-round Baron-Ferejohn bargaining with discounting ($\beta < 1$), the richer environment produces two screening cutoffs (one per round) and an entry threshold that creates a second channel through which the voting rule matters. Theorem~1 shows that the pattern identified here generalizes: conditional on entry, unanimity gives the hegemon a strictly higher payoff at every level of uncertainty. The institutional classification (Proposition~4) establishes that unanimity dominates wherever it can sustain entry; majority dominates only through wider institutional viability.

4 The Model

Definition 1 (Institutional Design Game)**.** *The game $\Gamma(N, p, r, \alpha, \beta, c)$ has the following primitives.*

- *There are $N \geq 3$ players: one hegemon H and $N-1$ weak states $W_1, \ldots, W_{N-1}$.*
- *Nature draws $\theta \in \{0, 1\}$ with prior* $\Pr(\theta = 1) = p \in (0, 1)$.
- *The state determines the value of multilateral cooperation: $V(0) = 1$ and $V(1) = r > 1$.*
- *The hegemon observes θ; the weak states do not.*
- *Disagreement payoffs: each W receives 0;[1] H receives $\alpha V(\theta)$, with $\alpha \in (0, 1/r)$.[2]*
- *The common discount factor is $\beta \in (0, 1)$.*
- *Each weak state that joins the institution pays an entry cost $c > 0$.*

[1] The normalization $d_W = 0$ is without loss of generality: it requires only that weak states' outside options are symmetric and strictly lower than the hegemon's. The pie $V(\theta)$ is then interpreted as the surplus above weak states' common outside option.

[2] The proportional form $d_H = \alpha V(\theta)$ captures bilateral alternatives that scale with the value of multilateral cooperation—the same factors that make multilateral cooperation valuable also improve the hegemon's outside option. The restriction $\alpha < 1/r$ ensures bilateral alternatives are strictly dominated by multilateral cooperation for all states of the world.

- *The voting rule is $R \in \{M, U\}$: under majority (M), decisions pass with $q = \lfloor N/2 \rfloor + 1$ votes; under unanimity (U), decisions require all N votes.*[3] *Both rules have symmetric proposal rights (each player recognized with probability $1/N$).*

Let $V_e(\mu) \equiv 1 + \mu(r-1)$ denote the expected value of cooperation at posterior belief μ, and define $x \equiv (N-1)\alpha r$ as a notational shorthand.

The two-round bargaining structure follows the Baron-Ferejohn framework that is standard in legislative bargaining. A single round would reduce the game to a one-shot ultimatum, and one might question whether the screening mechanism depends on that special structure. Two rounds ensure that rejection in Round 1 leads to a continuation game—not immediate disagreement—so the hegemon's reservation value in Round 1 reflects a credible outside option within the institution. The second round also generates off-path belief updating: when the high type rejects an aggressive offer in Round 1, weak states learn $\theta = 1$ with certainty, which pins down the Round 2 offers and, by backward induction, the Round 1 screening cutoff. Beyond robustness, the two-round structure generates analytically richer results: the R1 screening cutoff is independent of α (Proposition 2), and the two screening cutoffs (one per round) create a richer conditional payoff advantage for the hegemon. The motivating example (Section 3) shows that the mechanism may operate in one round; the two-round structure confirms that it survives in the richer Baron-Ferejohn environment.

The game has three stages.

1. **Stage 0 (Institutional choice).** The hegemon chooses the voting rule $R \in \{M, U\}$.
2. **Stage 1 (Entry).** Nature draws θ; the hegemon observes θ. Weak states hold prior belief $p = \Pr(\theta = 1)$ and simultaneously decide whether to enter (paying cost c). The hegemon participates automatically—it has chosen the institution at Stage 0 and faces no entry cost. The institution forms if and only if all $N-1$ weak states enter; partial entry is not permitted.[4] Since weak states are ex ante identical, I restrict attention to the symmetric entry equi-

[3] In this game, each player's action is binary: accept or reject. There is no abstention or silence option. The institutional distinction between consensus—where silence is treated as tacit approval—and formal unanimity—where explicit approval is required—therefore does not arise: with binary actions {accept, reject}, the two rules are equivalent. I use the terms interchangeably. In practice, "consensus" at the WTO refers to the norm of not calling a formal vote, which approximates unanimity when any member can block by objecting.

[4] The all-or-nothing entry technology is a simplification that avoids a separate extensive form for coalition formation at the entry stage. It is appropriate when the institution requires broad membership to function (as in a multilateral trade round) or when the voting rule is defined over a fixed membership. The parameter N should be read as the size of

librium: all $N-1$ weak states enter if and only if the expected equilibrium payoff satisfies $V_W^{R1}(p,R) \geq c$; otherwise no institution forms and each player receives their disagreement payoff.

3. **Stage 2 (Bargaining).** If the institution forms, members play a two-round Baron-Ferejohn bargaining game over a pie of size $V(\theta)$:
   - **Round 1**: a proposer is selected uniformly at random (probability $1/N$). The proposer offers a division. All members vote according to rule R. If accepted, payoffs are realized. If rejected, the game proceeds to Round 2 with payoffs discounted by β.
   - **Round 2 (terminal)**: a new proposer is selected uniformly at random. The proposer offers a division. All members vote. If accepted, payoffs are realized. If rejected, each player receives their disagreement payoff.

The solution concept is Perfect Bayesian Equilibrium (PBE). Strategies are sequentially rational given beliefs, and beliefs are updated by Bayes' rule on the equilibrium path. Off-path beliefs satisfy sequential rationality.[5]

Throughout the game, the hegemon's information sets are singletons: H knows θ at every decision node. Each weak state's information sets pool the two branches of θ: at any decision node, W_i knows the history of proposals and responses, but not θ. Weak states hold belief μ over θ, which equals the prior p at the entry stage and may be updated during bargaining (e.g., after a rejection in Round 1). This asymmetry is what generates the screening problem when a weak state proposes under unanimity.

Figures 1 and 2 present the extensive form of the game: Figure~1 covers institutional choice and entry (Stages 0–1); Figure~2 details the bargaining subgame under unanimity (Stage 2).

the institution the hegemon proposes to create, not the total number of states in the international system. The hegemon invites a specific group of $N-1$ weak states; the entry decision is whether to join this particular institution.

[5]Tie-breaking convention: when H is indifferent between accepting and rejecting an offer (offer equals reservation value), H accepts.

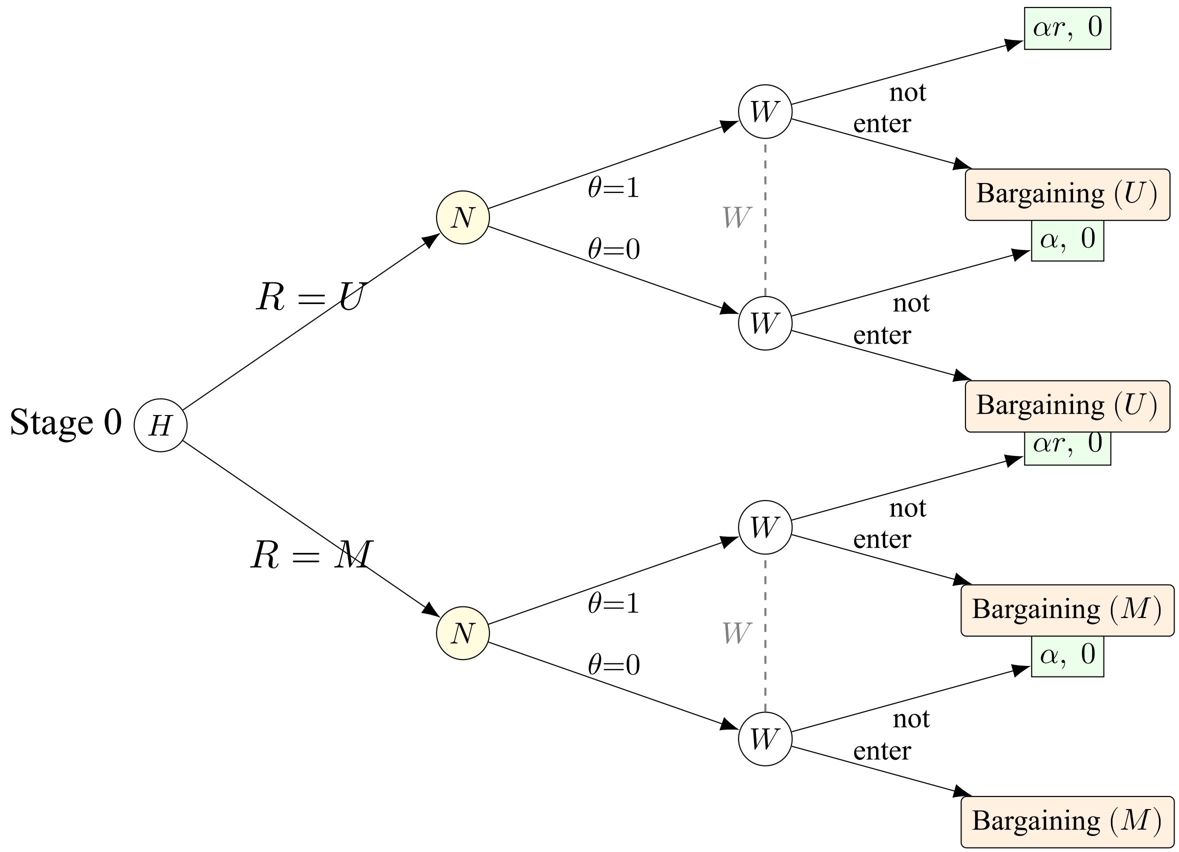

Figure 1: Extensive form: Stages 0–1 (institutional choice and entry). At Stage 0, H selects voting rule $R \in \{M, U\}$. Nature draws $\theta \in \{0, 1\}$; H observes θ, weak states hold prior $p = \Pr(\theta{=}1)$ and decide entry (cost c). Dashed lines: W's information sets (W does not observe θ). Non-entry payoffs: $(u_H, u_W) = (\alpha V(\theta),\ 0)$.

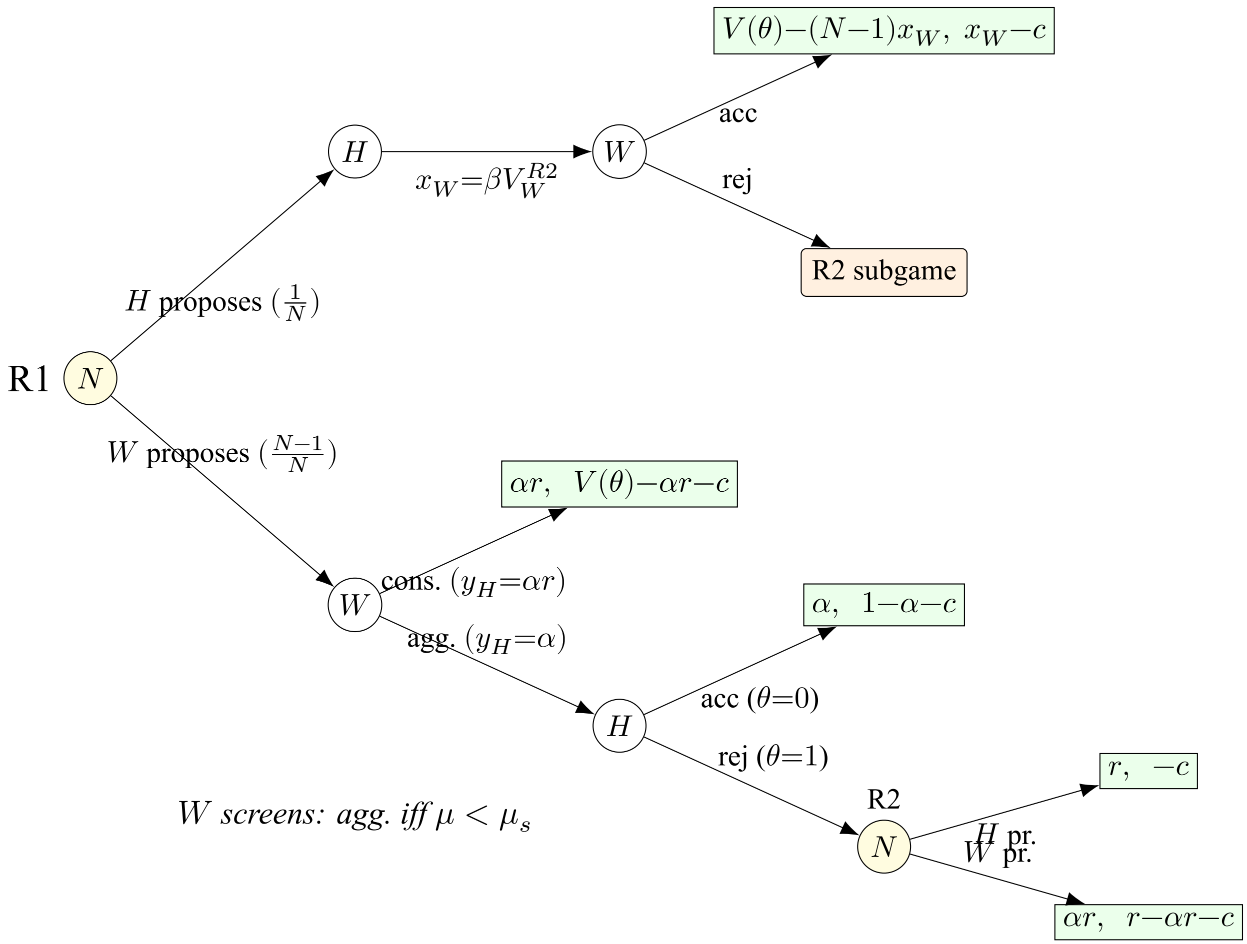

Figure 2: Extensive form: Stage 2 (bargaining under unanimity, Rounds 1–2). Payoffs are (u_H, u_W). A proposer is selected uniformly ($1/N$). When W proposes, W chooses between an aggressive offer that only $\theta=0$ accepts and a conservative offer that both types accept. Under the aggressive offer, $\theta=1$ rejects, revealing the state and leading to R2 (payoffs discounted by β). In R2, W knows $\theta=1$ (complete information); W offers αr, H offers 0; disagreement yields $(\alpha r, -c)$. When H proposes in R1, H offers βV_W^{R2} to each W; W accepts in equilibrium. Offer labels and terminal payoffs show R2 values (where H's reservation is $\alpha V(\theta)$) for readability; exact R1 offer levels are $\beta V_H^{R2}(\theta=0, \mu'=1)$ (aggressive) and $\beta V_H^{R2}(\theta=1)$ (conservative), derived in Appendix A.3. Under majority rule, H's vote is not pivotal, eliminating the screening mechanism (Proposition 1).

5 Majority Rule: Linear Bargaining Without Screening

Under majority rule, a proposal passes with $q = \lfloor N/2 \rfloor + 1$ votes. The key structural feature is that weak states can form a winning coalition without including the hegemon. When a weak state proposes, it assembles a majority from the other weak states alone; the hegemon, excluded from the coalition, captures its bilateral alternative $\alpha V(\theta)$ regardless of what weak states believe about the state of the world.

Proposition 1 (Majority rule produces no screening)**.** *Under majority rule with symmetric proposal rights, the hegemon's expected continuation value $E_\theta[V_H^{R1}(\theta, \mu, M)]$ is affine in posterior beliefs μ. There is no screening cutoff.*

Proof. See Appendix B.1. □

Because the hegemon's vote is never needed, weak states never face a screening problem, and the hegemon earns no screening rent. Majority transforms bargaining into coalition arithmetic: the proposer buys the cheapest votes, and the hegemon is simply paid its outside option.

6 Consensus: Screening Through Pivotal Inclusion

Under unanimity, every proposal requires the approval of all members, including the hegemon. When a weak state proposes, it must secure H's vote without knowing whether the pie is high or low. As in the motivating example (Section 3), this creates a screening problem: the proposer must choose between an aggressive offer that only the low type accepts and a conservative offer that both types accept. The conservative offer overpays the low type, which is the cost of uncertainty.

This screening logic operates in both bargaining rounds. In each round, the weak proposer's indifference between aggressive and conservative offers defines a screening cutoff: a belief threshold above which the proposer switches from aggressive to conservative treatment. The R2 cutoff has a closed form in all cases; the R1 cutoff has a closed form when the hegemon's outside option is moderate (Appendix A).

Proposition 2 (R1 screening cutoff under consensus)**.** *Under unanimity, there exists a unique screening cutoff $\mu_s^{R1} \in (0, 1)$ such that the weak proposer prefers the aggressive offer for $\mu < \mu_s^{R1}$*

and the conservative offer for $\mu > \mu_s^{R1}$. When $\alpha < \bar{\alpha}(r, \beta, N)$, the cutoff takes the closed form:

$$\mu_s^{R1} = \frac{\phi - \sqrt{\phi^2 - 4(N-2)}}{2(N-2)}, \qquad \phi \equiv \frac{rN - \beta(N - 1 + r)}{\beta(r-1)}, \tag{1}$$

and is independent of α. The regime threshold is

$$\bar{\alpha}(r, \beta, N) \equiv \frac{\mu_s^{R1} \cdot r}{r - 1 + \mu_s^{R1}}, \tag{2}$$

where μ_s^{R1} is evaluated from (1)*. For $\beta = 1$, the cutoff simplifies to $\mu_s^{R1} = 1/(N-2)$.*[6]

Proof. See Appendix B.2. □

The independence of the screening cutoff from the hegemon's outside option strength, which holds when $\alpha < \bar{\alpha}$—is substantively important. What determines how aggressively weak states treat the hegemon is not how valuable bilateral alternatives are, but how much the state of the world affects the gains from cooperation, that is, the dispersion $r - 1$. The cutoff also depends on bargaining patience and the number of states that must agree.

Proposition 3 (Screening creates an informational rent)**.** *Under unanimity, the hegemon's expected continuation payoff in R1 has an upward jump at μ_s^{R1}:*

$$\textit{Jump} = (1 - \mu_s^{R1}) \cdot \frac{(N-1)\beta(r-1)}{N^2} > 0. \tag{3}$$

Proof. See Appendix B.3. □

At the screening cutoff, weak proposers switch from treating the hegemon as the low type to hedging against the possibility of the high type. The conservative offer pays the low type as if it could be the high type, creating a discrete upward shift in the hegemon's expected payoff. The screening rent is largest when the dispersion in cooperation value is high (large r), when bargaining is patient enough to make the continuation threat credible (large β), and in smaller organizations, where the

[6] When $\alpha \geq \bar{\alpha}$, the R1 cutoff falls on the low R2 branch and depends on α (Appendix A.5). The cutoff remains unique and the screening jump remains positive. The proof of Theorem 1 (Appendix B.5) covers both regimes explicitly: the payoff decomposition is additive in the R1 and R2 corrections regardless of the relative ordering of μ_s^{R1} and μ_s^{R2}, and the endpoint argument establishes $D(\mu) > 0$ on all branches. The Corollary and Proposition then follow from Theorem 1 without further dependence on the regime.

per-member screening advantage $(N-1)/N^2$ is larger (the expression is monotonically decreasing in N for $N \geq 2$).

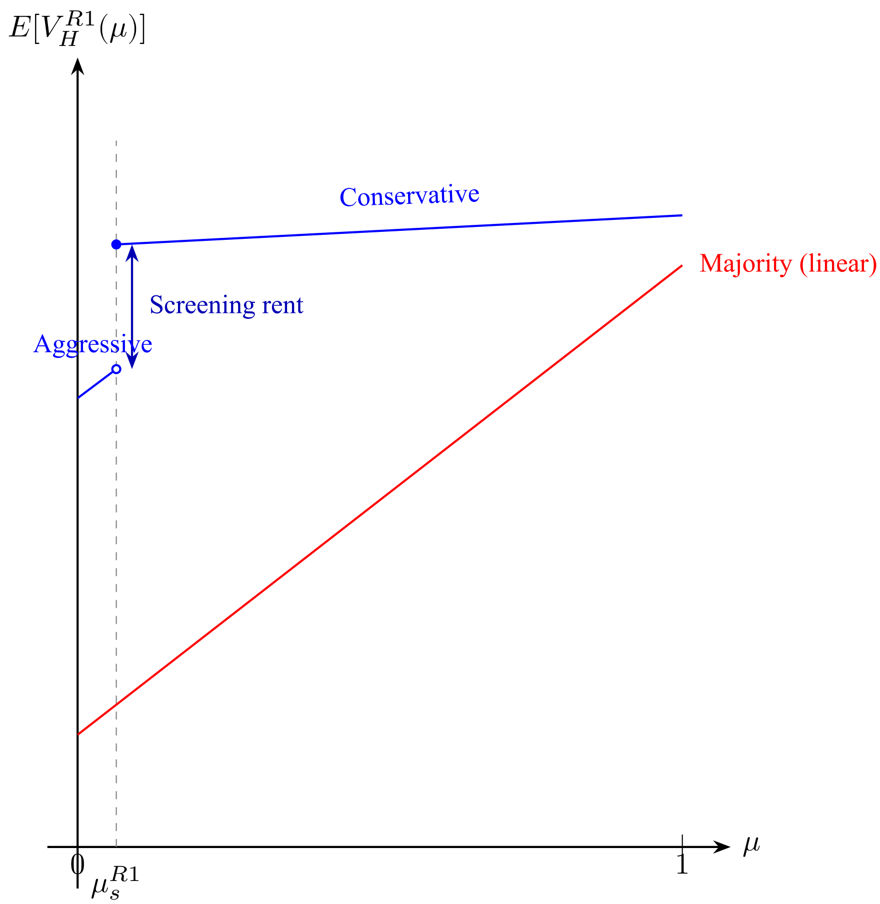

Figure 3: Schematic of the hegemon's expected R1 payoff as a function of posterior belief μ, based on Example 1 parameters ($N = 13$, $r = 1.5$, $\alpha = 0.19$, $\beta = 0.9$). Under majority (red), the payoff is linear in μ with no discontinuity. Under unanimity (blue), the payoff has two branches separated by an upward jump at the screening cutoff $\mu_s^{R1} \approx 0.064$: to the left, weak proposers play aggressively; to the right, they switch to conservative offers that overpay the low type. The gap is the screening rent that drives the hegemon's preference for unanimity (Theorem 1).

Example 1 (Screening jump: concrete magnitudes)**.** *Let* $N = 13$*,* $r = 1.5$*,* $\alpha = 0.19$*,* $\beta = 0.9$*.*[7]

[7] The parameters are calibrated to OPEC circa 1986 (Section 9.1). $N = 13$ is the number of OPEC members. $r = 1.5$ reflects the ratio of cartel-sustained to competitive oil prices ($12 to $8 per barrel in the 1986–87 period). $\alpha = 0.19$ is Saudi Arabia's outside-option share, computed as the difference between Saudi Arabia's market share in the non-cooperative equilibrium ($\approx 25\%$) and the average share of a remaining member ($\approx 6\%$), following the normalization $d_W = 0$ [@griffin1994oil]. $\beta = 0.9$ reflects moderate discounting, standard in Baron-Ferejohn

The R1 screening cutoff is $\mu_s^{R1} \approx 0.064$*. At beliefs just below the cutoff, weak proposers play aggressively and the hegemon's expected payoff is* $E[V_H^{R1}] \approx 0.315$*. Just above the cutoff, the payoff jumps to* ≈ 0.345*—an increase of about* 2.9% *of expected surplus. Under majority at the same belief,* $E[V_H^{R1}] \approx 0.234$*, with no jump anywhere. The screening mechanism gives the hegemon* 35% *more than majority on the aggressive branch and* 48% *more on the conservative branch. The discrete gap between branches is the screening rent that drives the hegemon's preference for unanimity.*

7 Entry and Institutional Viability

Having established that unanimity generates a conditional payoff advantage for the hegemon (Propositions 2–3), I now turn to the entry margin, the channel through which majority can nonetheless dominate.

A weak state enters the institution if its expected bargaining payoff exceeds the entry cost $c > 0$.[8] Because unanimity gives the hegemon a higher payoff conditional on participation (as shown in Section 8 below), it leaves less surplus for weak states, making entry harder: $\tau(U) \geq \tau(M)$.

When the institution does not form, the hegemon receives $\alpha V_e(\mu)$ from its bilateral alternative. Since this outside-option payoff is affine in μ and common to both rules, the institutional comparison depends only on the gain from bargaining relative to this baseline.

Definition 2 (Net gain function)**.** *The hegemon's* net gain *from the institution under rule* R *at posterior* μ *is*

$$v(\mu, R) \equiv \begin{cases} E_\theta[V_H^{R1}(\theta, \mu, R)] - \alpha V_e(\mu) & \textit{if weak states enter,} \\ 0 & \textit{otherwise.} \end{cases}$$

Under majority, $v(\mu, M)$ is affine above the entry threshold. Under unanimity, the screening jump

applications.

[8]The entry cost c may represent fixed costs of negotiating accession, per-round costs of preparing delegations, or political costs of committing to multilateral disciplines. In established organizations, c is best interpreted as the cost of *substantive engagement*: allocating scarce diplomatic and analytical resources to a particular negotiating round. Since $V(\theta)$ is normalized independently of N, per-capita payoffs scale as $O(1/N)$. For large-N applications, c should be read as a per-capita cost: writing $c = \tilde{c}/N$, the entry condition becomes $N \cdot V_W \geq \tilde{c}$, where $N \cdot V_W$ is $O(1)$. All equilibrium cutoffs (μ_s^{R1}, μ_s^{R2}, α^*) are homogeneous of degree zero in the surplus and thus independent of this normalization.

at μ_s^{R1} creates a discontinuity that gives the hegemon a higher net gain at every belief where entry occurs (Theorem~1).

8 Institutional Choice

This section establishes the paper's main result: conditions under which the hegemon prefers unanimity.

8.1 Conditional payoff comparison

Before turning to the full institutional comparison, I establish the paper's central result: whenever the hegemon's bilateral alternatives are not too valuable, unanimity gives the hegemon a strictly higher expected bargaining payoff than majority at every prior belief.

Define the *conditional dominance threshold*

$$\alpha^*(N,\beta) \equiv \frac{\beta(q-1)}{N(N-1)-\beta(N^2-N-q+1)}, \qquad q = \lfloor N/2 \rfloor + 1. \tag{4}$$

The parameter α measures the strength of the hegemon's bilateral outside option relative to multilateral cooperation. The threshold α^* marks the point at which the cost of buying unanimous agreement—$N-1$ votes, versus $q-1$ under majority—outweighs the screening advantage. When $\alpha < \alpha^*$, the screening rent from unanimity exceeds the higher vote-buying cost at every belief; when $\alpha \geq \alpha^*$, this fails near $\mu = 1$, where screening rents vanish and the cost difference is decisive.

Theorem 1 (Conditional payoff dominance of unanimity)**.** *The condition $\alpha < \alpha^*(N,\beta)$ holds if and only if, for every $\mu \in (0,1]$,*

$$E_\theta[V_H^{R1}(\theta,\mu,U)] > E_\theta[V_H^{R1}(\theta,\mu,M)].$$

Proof. See Appendix B.5. $\square$

The proof (Appendix B.5) proceeds by comparing the conditional payoff difference $D(\mu) \equiv$

$E[V_H(\mu, U)] - E[V_H(\mu, M)]$. Under majority, the hegemon's payoff is affine in beliefs: weak proposers can exclude the hegemon, so its private information creates no screening term. Under unanimity, the payoff difference decomposes additively into three components—a baseline that captures the cost of buying agreement under each rule, and two corrections that capture the screening rent when weak proposers switch to conservative offers and the continuation-value effect of screening in the second round. Each component is affine in μ on its domain, so positivity on any branch reduces to checking endpoints. The tightest case is $\mu = 1$, where both screening corrections vanish and the comparison reduces to vote-buying and outside-option terms alone. Evaluating $D(1) > 0$ yields exactly the condition $\alpha < \alpha^*(N, \beta)$. At all other branch endpoints, the baseline term is large enough to absorb any negative correction, so the affine structure implies $D(\mu) > 0$ for every $\mu \in (0, 1]$.

The condition is therefore pinned down by the no-uncertainty endpoint. At $\mu = 1$, the hegemon is known to be the high type, so unanimity can dominate majority only if its bargaining advantage exceeds the cost of buying all votes. The intuition for why the advantage holds throughout the belief space has two parts. When the hegemon proposes, unanimity depresses weak states' continuation values (because rejection leads to a game where the hegemon is pivotal), allowing the hegemon to buy agreement more cheaply. When a weak state proposes, unanimity forces the proposer to secure the hegemon's vote, generating the screening rent identified in Proposition 3. Both forces favor the hegemon.

Because unanimity's vote-buying cost grows with N while majority's grows more slowly, α^* decreases with organization size: for $\beta = 0.9$, $\alpha^* \approx 0.47$ when $N = 5$ but falls to ≈ 0.03 when $N = 164$. The condition is easy to satisfy in small institutions but demanding in large ones. For a WTO-sized organization, conditional dominance requires that the hegemon's bilateral alternatives capture only a small fraction of the multilateral surplus. This does not eliminate the mechanism—the screening jump persists regardless of α—but it restricts the domain over which unanimity dominates at *every* belief. Section 9.2 discusses what happens outside this domain.

Remark 1 (Robustness beyond α^*)**.** *When $\alpha \geq \alpha^*$, the conditional advantage of unanimity fails* *only near $\mu = 1$. On the conservative branch ($\mu > \mu_s^{R1}$), the payoff difference decomposes as*

$$D(\mu) = \underbrace{D(1)}_{\text{without info}} + (1-\mu) \cdot \underbrace{\Gamma}_{\text{value of uncertainty}},$$

where $C_{buy} \equiv \beta(q-1)(1-\alpha)$ is the vote-buying cost under majority, $C_{out} \equiv N(N-1)\alpha$ is the aggregate outside-option cost, $D(1) = r[C_{buy} - C_{out}(1-\beta)]/N^2 \leq 0$ is the payoff comparison when the hegemon's type is known, and $\Gamma = (r-1)[C_{out} - C_{buy} + (N-1)\beta]/N^2 > 0$ is the marginal value of uncertainty for unanimity. The belief threshold above which majority dominates is $\bar{\mu} = 1 - |D(1)|/\Gamma$. Numerically, $\bar{\mu}$ is remarkably stable: for $N = 164$ and $\beta = 0.9$, increasing α from 0.05 to 0.49 (an 18-fold increase beyond α^*) only lowers $\bar{\mu}$ from 0.87 to 0.71 when $r = 1.5$. The mechanism fails only near certainty about the magnitude of the gains from cooperation—precisely where private information has least value (Figure 5 in Section 9.2 maps the full (α, μ) frontier).

8.2 Formation sets and institutional comparison

Let $\mathcal{F}_R \equiv \{p \in (0,1] : V_W^{R1}(p,R) \geq c\}$ denote the *formation set* under rule R: the set of priors at which the institution forms. Since weak states' aggregate payoff is weakly lower under unanimity whenever the hegemon's payoff is higher (Theorem~1), the unanimity entry constraint is harder to satisfy; hence $\mathcal{F}_U \subseteq \mathcal{F}_M$ (Appendix B.6). The institutional comparison is therefore determined by the entry margin: unanimity dominates when the institution forms, but majority can dominate when only it can induce entry.

Corollary 1 (Unanimity dominance on the formation set)**.** *Suppose $\alpha < \alpha^*(N, \beta)$. For every prior $p \in \mathcal{F}_U$,*

$$V_H(U,p) > V_H(M,p).$$

Moreover, $\mathcal{F}_U \subseteq \mathcal{F}_M$.

Proof. See Appendix B.6. □

The argument combines budget balance with Theorem 1. Under majority, agreement occurs without delay and the pie is divided exactly among N players, so the hegemon's and weak states' expected payoffs sum to $V_e(\mu)$. Under unanimity, rejection on the aggressive branch destroys surplus through discounting, so the same budget constraint holds as a weak inequality. Since the hegemon captures strictly more under unanimity (Theorem 1), weak states must receive strictly less at every belief. The entry threshold is therefore higher under unanimity, and any prior that sustains entry under unanimity also sustains it under majority.

Remark 2 (Weak states prefer majority)**.** *Under the conditions of Theorem 1, $V_W(\mu, M) > V_W(\mu, U)$ for every $\mu \in (0, 1]$, and $\mathcal{F}_U \subseteq \mathcal{F}_M$. Unanimity redistributes from weak states to the hegemon; it is never Pareto-improving conditional on entry.*

Why, then, would weak states participate under unanimity? Because the hegemon chooses the voting rule. A weak state that refuses to enter under unanimity does not get majority instead—it gets no institution at all, forfeiting whatever surplus the organization would generate. Since weak states are ex ante identical and decide simultaneously, the symmetric equilibrium is all-or-nothing: either all enter or none does (Section 4). There is no scope for individual holdout to force a rule change. Conditional on the hegemon's choice of unanimity, entry is individually rational whenever $V_W(p, U) \geq c$. The irony is that the veto, which guarantees each weak state inclusion in every agreement, is precisely what creates the screening problem that benefits the hegemon: by guaranteeing inclusion, unanimity forces weak proposers to make offers without knowing the hegemon's type, generating the overpayment that drives the conditional payoff advantage.

The following result completes the institutional comparison by classifying all priors.

Proposition 4 (Institutional classification)**.** *Suppose $\alpha < \alpha^*(N, \beta)$. The hegemon's preferred rule is characterized as follows:*

$$\begin{aligned} U \succ M &\quad \textit{if and only if} \quad p \in \mathcal{F}_U, \\ M \succ U &\quad \textit{if and only if} \quad p \in \mathcal{F}_M \setminus \mathcal{F}_U, \\ U \sim M &\quad \textit{if and only if} \quad p \notin \mathcal{F}_M. \end{aligned}$$

Since $\mathcal{F}_U \subseteq \mathcal{F}_M$, majority's only advantage is that it can support institutional entry at priors for which unanimity cannot.

Proof. See Appendix B.8. □

When the prior is such that weak states find it individually rational to enter under unanimity, the hegemon strictly prefers it (Corollary~1). When only majority can sustain entry, majority dominates through wider institutional viability rather than better bargaining terms. When neither rule can induce entry, the voting rule is irrelevant. When entry costs are low enough that $\mathcal{F}_U = (0, 1]$, unanimity dominates at every prior.

Remark 3 (Supermajority, weighted voting, and strategic inclusion)**.** *The qualitative result is robust to the choice of majority threshold. Under any unweighted supermajority quota $q < N$, the exclusion logic is unchanged: weak proposers can still assemble a winning coalition without the hegemon, and no screening occurs. A higher quota reduces λ_M (because the hegemon must buy more coalition partners when it proposes), which has two consequences. First, the hegemon's payoff under majority falls, so the dominance threshold α^* increases with q: the condition $\alpha < \alpha^*$ becomes easier to satisfy. Second, by budget balance, weak states' payoff under majority rises, expanding the majority entry set $\mathcal{F}_M$. The unanimity entry set $\mathcal{F}_U$ is unaffected, since unanimity payoffs do not depend on the majority quota. In the (p, c) plane of Figure reffig:parameter-regions, this means the orange region (majority dominates via entry) expands at the expense of the white region (no institution forms), while the blue region ($\mathcal{F}_U$) remains fixed. Supermajority makes majority more accessible to weak states, but also makes it less attractive to the hegemon relative to unanimity, without altering the qualitative structure of the comparison.*

The more interesting extension is weighted voting. Consider weighted majority with weight $w_H > 1$ for the hegemon and weight 1 for each weak state, where the total weak-state weight still suffices to pass proposals without the hegemon. When w_H is small, including the hegemon in a winning coalition costs more than buying w_H additional weak states: $V_H^{R2}(\theta) > w_H \cdot V_W^{R2}(\mu)$ for all (θ, μ). In this regime, weak proposers exclude the hegemon exactly as under simple majority, and the payoff remains $\lambda_M(w_H) \cdot V_e(\mu)$, with λ_M increasing in w_H because the hegemon needs fewer coalition partners when it proposes. No screening occurs. However, if w_H exceeds a threshold w_H^* at which buying the hegemon's vote becomes cheaper than assembling an equivalent coalition of weak states, weak proposers rationally include the hegemon even though its vote is not formally required. At that point, the proposer faces the same screening problem as under unanimity: choosing between *an aggressive offer accepted by all types and a conservative offer rejected by* $\theta = 1$*. Screening thus depends on* strategic inclusion*, not on formal pivotality. One may thus interpret the unanimity–majority comparison in this paper as two polar cases of this continuum.*

Remark 4 (Information design)**.** *The model also clarifies why information design would matter differently under the two voting rules. Under majority rule, the hegemon's continuation payoff is affine in weak states' posterior belief once entry is feasible. Splitting beliefs therefore does not create additional value beyond the payoff implied by the prior. Under unanimity, by contrast, weak states must include the hegemon in any agreement. This creates a screening discontinuity at the posterior threshold at which entry becomes feasible. As in standard Bayesian persuasion [@kamenica2011bayesian], such nonconcavities are precisely where information design can matter. Thus, although persuasion is not necessary for the paper's main result, the model identifies why consensus institutions are more hospitable to informational leverage than majority institutions.*

Example 2 (The full institutional comparison)**.** *Continuing with the parameters from Example 1 (*$N = 13$*,* $r = 1.5$*,* $\alpha = 0.19$*,* $\beta = 0.9$*), suppose entry costs* $c = 0.07$*. Under majority, the institution forms for* $p \gtrsim 0.17$*:* $\mathcal{F}_M \approx [0.17, 1]$*. Under unanimity, the screening mechanism reduces weak states' surplus, so the institution forms only for* $p \gtrsim 0.38$*:* $\mathcal{F}_U \approx [0.38, 1]$*.*

For $p \in \mathcal{F}_U$ *(above* ≈ 0.38*), unanimity strictly dominates: at* $p = 0.50$*, the hegemon's payoff under unanimity exceeds majority by* 23%*. For* $p \in \mathcal{F}_M \setminus \mathcal{F}_U$ *(between* ≈ 0.17 *and* ≈ 0.38*), majority dominates because only it can sustain entry. The advantage of majority at pessimistic priors is entirely through the entry channel: conditional on participation, unanimity always gives the hegemon more (Theorem 1).*

8.3 Numerical characterization

Figure 4 maps the institutional classification directly. For each combination of prior p and entry cost c, the figure shows which rule the hegemon prefers: unanimity (blue), majority through the entry channel (orange), or neither (gray). The unanimity-dominance region is always nested inside the majority-entry region, confirming that $\mathcal{F}_U \subseteq \mathcal{F}_M$. The unanimity-dominance region expands with r (larger screening rents from greater type dispersion) and β (more patient bargaining strengthens the off-path threat that sustains screening), and contracts with α (stronger outside options raise the cost

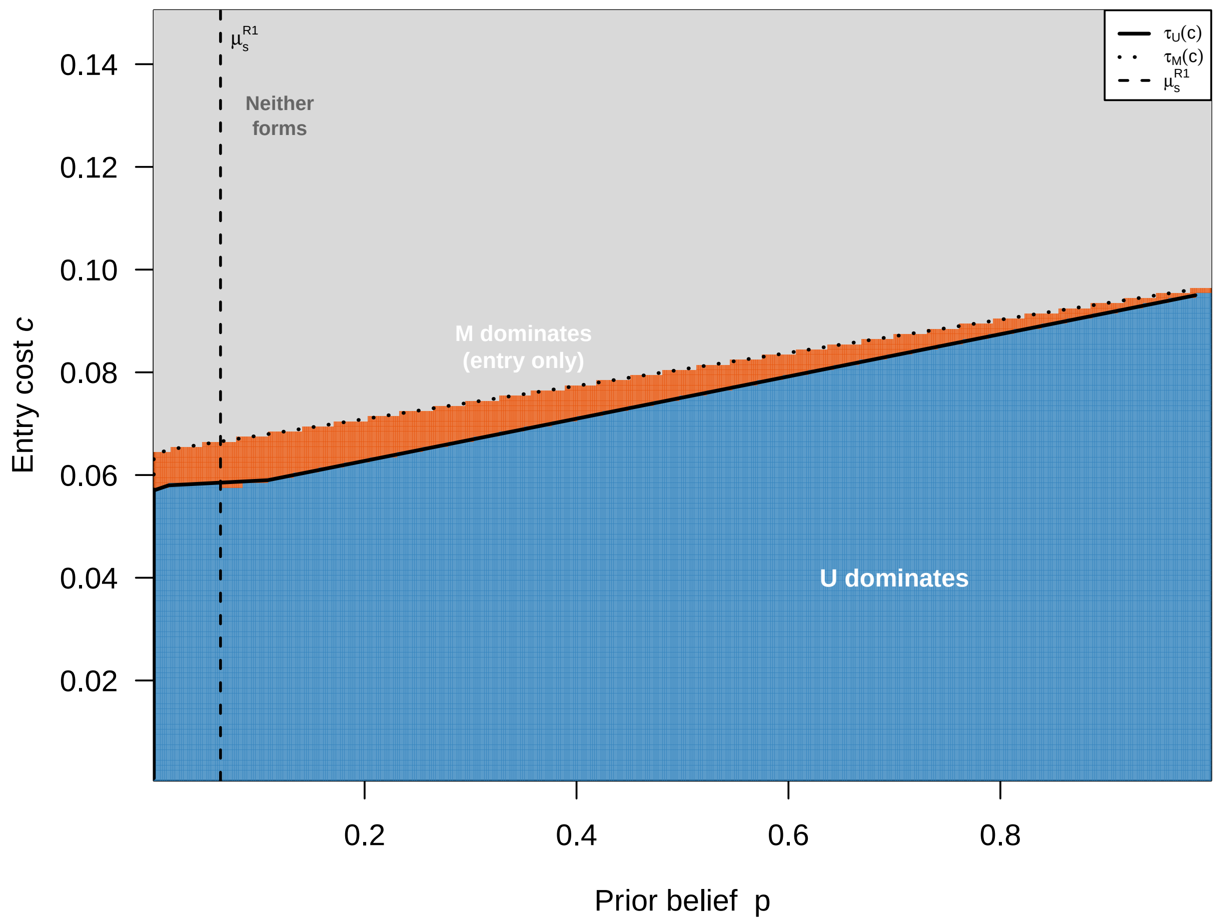

Figure 4: Institutional classification in the (p, c) plane ($N = 13$, $r = 1.5$, $\alpha = 0.19$, $\beta = 0.9$). Blue indicates pairs for which unanimity sustains entry and therefore dominates majority conditional on participation. Orange indicates pairs for which majority sustains entry but unanimity does not, so majority dominates through the entry channel. Gray indicates pairs for which neither rule sustains entry. The vertical dashed line marks the screening cutoff μ_s^{R1}. At this cutoff, weak proposers switch from aggressive to conservative offers under unanimity, generating an upward jump in the hegemon's payoff and, by budget balance, a downward jump in weak-state payoffs. This can make the unanimity formation set disconnected.

of buying unanimous agreement) and c (higher entry costs shrink $\mathcal{F}_U$ faster than $\mathcal{F}_M$, widening majority's scope advantage). The figure also illustrates why entry under unanimity need not be monotone in the prior. The same screening rent that gives the hegemon a conditional advantage under unanimity lowers weak-state payoffs at the cutoff, creating a local failure of entry under unanimity even when entry is feasible at lower and higher priors.

9 Discussion

9.1 Illustration: OPEC and the 1985–86 Price War

OPEC illustrates the screening mechanism in a setting where the institutional mapping is unusually clean. The hegemon is Saudi Arabia, the dominant producer with approximately 60% of global spare capacity. The weak states are the remaining OPEC members—Iran, Iraq, Nigeria, Venezuela, and others—who collectively depend on Saudi restraint to sustain cartel rents. The cooperation surplus $V(\theta)$ is the monopoly rent from collectively restricting supply above the competitive level. The state of the world θ is Saudi Arabia's true spare production capacity: how much additional oil Saudi Arabia can bring to market within 90 days. This capacity determines the value of the cartel agreement for all members, because it governs the credibility and costliness of the punishment that follows a breakdown. Saudi Arabia's outside option α is unilateral production—selling at market prices through bilateral contracts with China, India, and other importers—which yields substantial revenue even without OPEC coordination, given Saudi extraction costs of approximately \$3 per barrel, the lowest in the world (Alhajji and Huettner 2000; Nakov and Nuño 2013). The decision-making rule is unanimity: Article 11 of the OPEC Statute requires that "all decisions of the Conference, other than on procedural matters, shall require the unanimous agreement of all Full Members.' '[9]

The informational asymmetry at the core of the model is well documented. Saudi Arabia's true spare capacity has been closely guarded since 1982, when Oil Minister Yamani discontinued field-by-field production reporting to industry periodicals (Simmons 2005). No independent audit of

[9] Appendix C extends the model to continuous types. The screening rent is strictly positive for any continuous distribution on $[1, r]$ with full support, and for the uniform distribution the dominance threshold satisfies $\alpha^*_{\text{cont}} \geq \alpha^*$, suggesting that the binary model is the most conservative specification for the mechanism.

Saudi reserves was conducted until 2019, when DeGolyer & MacNaughton was engaged ahead of the Aramco IPO. Expert disagreement about actual deployable capacity is substantial: the IEA estimates Saudi spare capacity at approximately 2.4 million barrels per day, while independent analysts place the figure between 600,000 and 1 million—a divergence of 40–80% (Fattouh and Mahadeva 2013). In December 2025, the U.S. Energy Information Administration revised its methodology to distinguish theoretical'' fromeffective’’ production capacity, implicitly acknowledging that much of OPEC’s reported spare capacity may exist only on paper. Other OPEC members cannot independently verify Saudi claims, and this uncertainty is precisely what gives Saudi Arabia informational leverage: when weaker members must decide how much to concede in quota negotiations, they do not know how costly a Saudi-led price war would actually be.

The December 1986 OPEC agreement illustrates the screening mechanism. By mid-1985, endemic quota cheating by other members had driven Saudi Arabia’s production share to roughly 14% of OPEC output, and its revenue had fallen below the Cournot equilibrium payoff—the level it could guarantee unilaterally by producing at capacity without any cartel agreement (Griffin and Neilson 1994). The price collapse of 1985–86, in which spot prices fell from $28 to $8.55 per barrel, restored the conditions for cooperation by making the cost of non-cooperation vivid for all members (Griffin and Neilson 1994; Yergin 1991).

With cooperation reestablished under OPEC’s unanimity rule, Saudi Arabia proposed the terms of a new agreement in December 1986: a price target of $18 per barrel, a production ceiling of 15.8 million barrels per day, and a guaranteed quota share of approximately 25% for Saudi Arabia—nearly double its mid-1985 share. The other members accepted and, for the first time, began to comply with quotas in practice. This maps onto the branch of the model where the hegemon proposes in Round 1. The rent extraction does not come from the proposal itself—in a single-round ultimatum, any proposer captures the entire surplus. It comes from the continuation game: rejecting the hegemon’s offer leads to further bargaining rounds where weak states face the screening problem created by uncertainty about Saudi Arabia’s true spare capacity. This continuation threat is what depresses weak states’ reservation value under unanimity relative to majority. Members did not know how long Saudi Arabia could sustain low prices or how costly renewed confrontation would be. This uncertainty depressed their effective reservation value, allowing Saudi Arabia to secure a larger share than a transparent bargaining environment would have produced. Under a hypothetical majority rule, other members could have set quotas without Saudi consent, bypassing the screening problem entirely—but they would also have lost the enforcement mechanism that only Saudi spare capacity provides.

Recent events reinforce this interpretation. Angola left OPEC in January 2024 after its quota was cut by 350,000 barrels per day; its oil minister stated that OPEC was “not serving the country's interests.' ' The UAE exited in April 2026, citing a persistent gap between its production capacity of approximately 5 million barrels per day and its assigned quota of 3.2 million. Both departures are consistent with the model's institutional classification (Proposition~4): as a member's outside option improves—Angola developing non-oil sectors, the UAE investing in capacity expansion—the formation set $\mathcal{F}_U$ contracts, and unanimity ceases to be individually rational. OPEC's response—a 2027 reform commissioning independent capacity audits by DeGolyer & MacNaughton for 19 member states—is a direct attempt to reduce the informational asymmetry that sustains screening rents. If successful, it would narrow the gap between aggressive and conservative offers, reducing Saudi Arabia's informational advantage but potentially stabilizing the coalition by making quota allocation more transparent.

The OPEC case has limitations as an illustration of the static model. The repeated-game structure matters: Saudi Arabia's credibility in 1986 depended on its demonstrated willingness to sustain a price war, and this reputation has shaped subsequent negotiations (Griffin and Neilson 1994). The model captures the structure of each bargaining episode but not the reputational dynamics that link them. Weak states within OPEC are heterogeneous—Venezuela's fiscal breakeven exceeds \$100 per barrel while Kuwait's is approximately \$82—whereas the model assumes symmetric weak states. Saudi Arabia did not choose unanimity at OPEC's founding in 1960, when Venezuela was the largest producer; the unanimity rule reflected collective sovereignty concerns of all five founders. The model's prediction that unanimity benefits the hegemon applies conditional on the rule being in place, not as a claim about the founding moment.

9.2 Scope

Why symmetric proposal rights? Under either voting rule, if the hegemon were the exclusive proposer and weak states had no outside option from bargaining ($d_W = 0$), weak states' expected bargaining payoff would be zero. No weak state would pay an entry cost $c > 0$, and the institution would never form. Monopoly proposal power is self-defeating when participation is voluntary. The relevant institutional choice is therefore between voting rules, holding proposal rights symmetric.

What exactly is being compared? The institutional comparison is between unanimity and symmetric simple majority: both rules share the same random-proposer protocol, and the only difference is the voting threshold. In particular, majority rule here does not grant the hegemon agenda control, weighted voting, or privileged coalition-building capacity. This is a deliberate choice: the comparison isolates the effect of the voting rule on information use, holding all other institutional features constant. In practice, majority rules in international organizations often come bundled with weighted voting or asymmetric proposal rights—as in the IMF or World Bank—which would independently benefit the hegemon. The result should therefore be read as identifying a mechanism that operates through the voting rule alone, not as a claim that unanimity dominates all possible alternatives to symmetric majority. Remark 3 explores what happens when the hegemon's voting weight increases: the mechanism activates whenever the hegemon is strategically included in winning coalitions, whether or not its vote is formally required.

Why not compare consensus with hegemonic agenda control? Because agenda control combined with any participation constraint eliminates the institution itself. The puzzle is not why a hegemon does not impose dictatorship, but why it would choose a rule that makes its own approval necessary when majority would allow others to bypass it.

Why all-or-nothing entry? The all-or-nothing entry technology isolates the screening mechanism by suppressing dynamics that would arise under sequential accession—such as free-riding, learning from early joiners, or collective action problems among weak states. This simplification is most plausible for organizations where membership is negotiated as a package—founding moments like the GATT (1947), the WTO (1995), or the establishment of new regulatory bodies—rather than for organizations with rolling accession. With sequential entry, multiple mechanisms would operate in opposing directions, and the net effect on the hegemon's institutional preference would depend on their relative magnitudes. The present model focuses on the screening channel alone; characterizing its interaction with sequential entry dynamics is left open.

Why binary types? Binary types isolate the screening mechanism most transparently. The screening rent under unanimity is strictly positive for any continuous distribution with full support, while majority eliminates it entirely (Appendix C). For the uniform distribution, the dominance threshold satisfies $\alpha^*_{\text{cont}} \geq \alpha^*$, suggesting that binary types are the most conservative specification for the mechanism. Binary types therefore provide a lower bound on the screening advantage.

When consensus favors the hegemon. The mechanism requires that: (i) informational asymmetry is relevant (H knows θ, W does not); (ii) the hegemon has a valuable outside option ($\alpha > 0$); (iii) entry is costly and belief-dependent; (iv) the voting rule makes H pivotal.

Figure 5 maps the conditional dominance region in the (α, μ) plane. For each parameter combination, the blue region indicates beliefs at which unanimity gives the hegemon a higher continuation payoff; the red region indicates beliefs at which majority dominates. The solid curve traces the frontier $\bar{\mu}(\alpha)$ from Remark 1. Two patterns stand out. First, the frontier is remarkably flat: increasing α well beyond α^* shifts $\bar{\mu}$ only modestly downward, confirming that the screening advantage is robust across most of the belief space. Second, the red region is always concentrated near $\mu = 1$—near certainty about the magnitude of the gains from cooperation—where informational asymmetry is irrelevant and the cost of buying unanimous agreement dominates.

When majority dominates. The biconditional in Theorem 1 implies that $\alpha < \alpha^*$ is the largest parametric region where majority can never dominate through a higher conditional-on-entry payoff. Any advantage of majority must come through the entry margin: $\tau(U) > \tau(M)$. This is more likely when entry costs are high (creating a large gap where the institution forms only under majority), when α approaches α^* (weakening the conditional dominance), or when information is approximately public. For $\alpha \geq \alpha^*$, majority can also dominate through conditional bargaining payoffs near $\mu = 1$, because unanimity's cost of buying $N - 1$ votes outweighs the screening advantage.

The threshold α^* decreases with N because unanimity requires buying $N-1$ votes, while majority requires only $q-1$. But α^* governs dominance at *every* belief; the hegemon only needs unanimity to dominate at the actual prior. Even when α far exceeds α^*, unanimity dominates over most of the

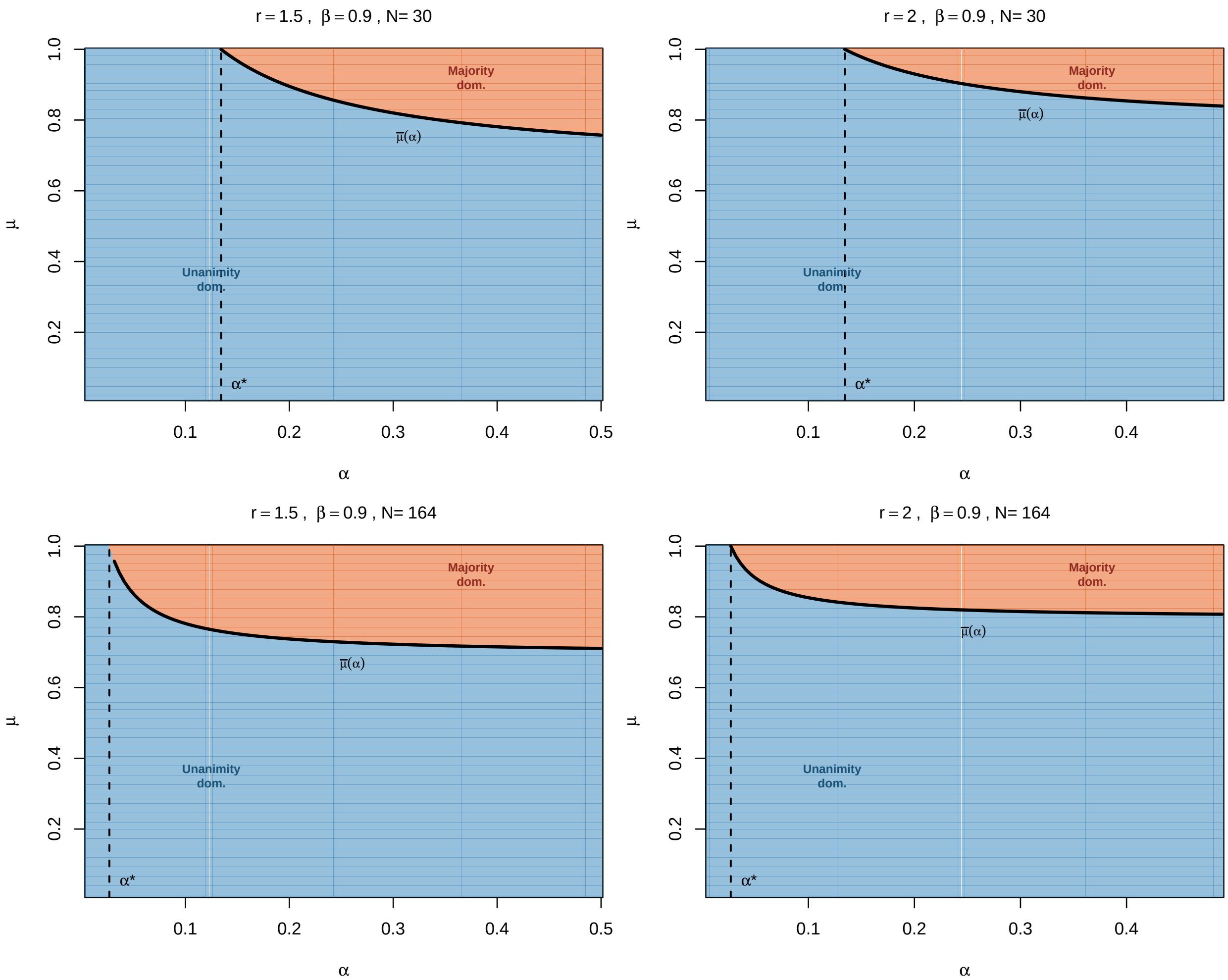

Figure 5: Conditional payoff dominance in the (α, μ) plane. Blue: unanimity dominates ($D(\mu) > 0$). Red: majority dominates ($D(\mu) < 0$). Solid curve: belief threshold $\bar{\mu}(\alpha)$ above which majority dominates (Remark~1). Dashed line: α^*. Left column: $r = 1.5$. Right column: $r = 2$. Top row: $N = 30$. Bottom row: $N = 164$. All panels: $\beta = 0.9$.

belief space: majority's conditional advantage is confined to a narrow region near $\mu = 1$, where the hegemon's type is nearly known and informational asymmetry is irrelevant. As Remark 1 shows, increasing α from 0.05 to 0.49 for $N = 164$ only shifts the frontier from $\bar{\mu} = 0.87$ to 0.71—an 18-fold increase in the outside option compresses the unanimity-dominance region by less than 20%. Figure 5 illustrates: even in the $N = 164$ panels, the blue region—where unanimity gives the hegemon a higher conditional payoff—covers the large majority of the (α, μ) plane, while the red region is always concentrated in the corner near certainty.

The biconditional in Theorem 1 implies that this boundary is sharp. At $\alpha = \alpha^*$, the dominance difference $D(\mu)$ equals zero exactly at $\mu = 1$; for $\alpha > \alpha^*$, majority strictly dominates near $\mu = 1$ (where the vote-buying cost is decisive), while unanimity can still dominate at lower beliefs. The paper's main results are stated for $\alpha < \alpha^*$, which ensures a clean separation between the conditional and entry channels.

10 Conclusion

This paper has identified conditions under which consensus is a rational choice for a hegemon. The key is to distinguish between two modes of exercising power under different voting rules. Majority rule with symmetric proposals lets weak states bypass the hegemon, eliminating the screening problem. Unanimity forces weak states to include the hegemon under uncertainty, creating a screening cutoff that gives the hegemon a conditional payoff advantage at every level of uncertainty. The institutional comparison has a clean structure: unanimity dominates wherever it can sustain entry, and majority's only advantage is wider institutional viability.

The argument does not imply that all consensus institutions are designed for this reason, or that informal power disappears under consensus. It shows something more specific: under identifiable conditions on informational asymmetry and the prospects of cooperation, unanimity becomes the preferred institutional arrangement for a hegemon. Consensus may be less a concession by power than a way of exercising power through a different channel—the channel of information.

The main results use a binary type space. It is important to distinguish what generalizes from what does not. The screening rent under unanimity is strictly positive for any continuous distribution on $[1, r]$ with full support (Appendix C). Under majority, W bypasses H, and the rent is zero. This structural contrast—screening under unanimity, no screening under majority—is distribution-free. What does *not* generalize without qualification is the *dominance threshold* α^*. For binary types, Theorem 1 provides a sharp biconditional. For the uniform distribution on continuous types, the dominance threshold α^*_{cont} is weakly larger than α^*, suggesting that binary types are the most conservative case. For other distributions, the threshold has not been characterized. A general characterization of α^* for arbitrary distributions remains an open question.

Several extensions remain. First, if weaker states can invest in independent analytical capacity, informational power erodes endogenously over time. Second, heterogeneous outside options across weak states (middle powers) would enrich the screening dynamics. Third, repeated play within an established consensus institution could generate learning that narrows the informational gap. These directions point toward a richer theory of how informational power rises and declines within institutional frameworks.

Appendix A: Bargaining Derivations

Notation summary

Symbol	Meaning	Defined in
N	Number of players (1 hegemon $+N-1$ weak states)	Def. 1
$\theta \in \{0,1\}$	State of the world	Def. 1
p	Prior $\Pr(\theta = 1)$	Def. 1
μ	Posterior belief $\Pr(\theta = 1 \mid \text{history})$; equals p at entry, updated after R1 rejection	Sec. 3
$V(\theta)$	Value of cooperation: $V(0) = 1, V(1) = r$	Def. 1
$r > 1$	High-type value	Def. 1
$\alpha \in (0, 1/r)$	Outside-option share: $d_H = \alpha V(\theta)$	Def. 1
$\beta \in (0,1)$	Common discount factor	Def. 1
$c > 0$	Entry cost per weak state	Def. 1
$V_e(\mu)$	Expected cooperation value $1 + \mu(r-1)$	Sec. 3
x	Shorthand $(N-1)\alpha r$	Sec. 3
q	Majority quota $\lfloor N/2 \rfloor + 1$	Def. 1
$R \in \{M, U\}$	Voting rule (majority / unanimity)	Def. 1
μ_s^{R2}	R2 screening cutoff	App. A.2
μ_s^{R1}	R1 screening cutoff	Prop. 2
ϕ	Auxiliary for R1 cutoff: $[rN - \beta(N{-}1{+}r)]/[\beta(r{-}1)]$	Prop. 2
$\tau(R)$	Entry threshold under rule R	Sec. 6
$v(\mu, R)$	Hegemon's net gain from institution: $E[V_H^{R1}] - \alpha V_e(\mu)$ if entry, 0 otherwise	Sec. 6
$\bar{\alpha}$	Regime threshold for α-independent R1 cutoff	Prop. 2
α^*	Threshold for conditional dominance (Thm. 1): necessary and sufficient	Eq. (4)
$\omega(\mu)$	R2 continuation shorthand: $(N{-}2)\beta V_W^{R2}(\mu)$	App. A.3
κ_M	Majority weak-state coefficient: $(1{-}\alpha)[N(N{-}1){+}\beta(q{-}1)]/(N^2(N{-}1))$	App. A.7
$D(\mu)$	Conditional payoff difference $V_H^{R1}(\mu, U) - V_H^{R1}(\mu, M)$	Thm. 1
Γ	Marginal value of uncertainty for unanimity	Remark 1
$\bar{\mu}$	Belief threshold above which majority dominates: $1 - \lvert D(1)\rvert/\Gamma$	Remark 1
$C_{\text{buy}}, C_{\text{out}}$	Vote-buying cost, outside-option cost (proof)	App. B.5
λ_M	Majority payoff coefficient	App. B.1
$\mathcal{F}_R$	Formation set under rule R: $\{p : V_W^{R1}(p, R) \geq c\}$	Sec. 7

A.1 Majority: R2 and R1

Under majority, weak proposers form a winning coalition from other weak states, excluding H. The hegemon captures $\alpha V(\theta)$ from its bilateral alternative. H's payoff is type-specific and does not depend on W's beliefs.

R2 continuation values:

$$V_H^{R2}(\theta, M) = \frac{V(\theta)[1+(N-1)\alpha]}{N} \tag{5}$$

$$V_W^{R2}(\mu, M) = \frac{(1-\alpha)V_e(\mu)}{N} \tag{6}$$

R1 values follow by backward induction with the same structure (no screening, linear).

A.2 Unanimity: R2

When H proposes: offers 0 to all W's, keeps $V(\theta)$.

When W proposes: the screening subgame has cutoff $\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$. Below it, W plays aggressive ($y_H = \alpha$); above, conservative ($y_H = \alpha r$).

The hegemon's R2 continuation values under unanimity are:

$$V_H^{R2}(\theta = 1, \mu) = \frac{r[1+(N-1)\alpha]}{N} \quad \text{(constant in } \mu\text{)}, \tag{7}$$

$$V_H^{R2}(\theta = 0, \mu < \mu_s^{R2}) = \frac{1+(N-1)\alpha}{N}, \tag{8}$$

$$V_H^{R2}(\theta = 0, \mu > \mu_s^{R2}) = \frac{1+(N-1)\alpha r}{N} \quad \text{(overpaid)}. \tag{9}$$

Weak states' R2 continuation values under unanimity:

$$V_W^{R2}(\mu < \mu_s^{R2}) = \frac{(1-\mu)(1-\alpha)}{N} \tag{10}$$

$$V_W^{R2}(\mu > \mu_s^{R2}) = \frac{V_e(\mu) - \alpha r}{N} \tag{11}$$

Continuity at μ_s^{R2} is verified by direct substitution.

A.3 Unanimity: R1 screening derivation

The proposing W compares:

$$F_1^{con}(\mu) = V_e(\mu) - \frac{\beta(r+x)}{N} - \omega(\mu) \tag{12}$$

$$F_1^{agg}(\mu) = (1-\mu)\left[1 - \frac{\beta(1+x)}{N} - \omega(\mu)\right] + \frac{\mu\beta r(1-\alpha)}{N} \tag{13}$$

where $\omega(\mu) = (N-2)\beta V_W^{R2}(\mu)$.

The difference $\Delta_1(\mu) = F_1^{agg} - F_1^{con}$ is piecewise quadratic. On the high branch ($\mu > \mu_s^{R2}$), the α-dependent terms cancel, yielding the quadratic:

$$(N-2)\beta(r-1)\mu^2 + [\beta(N-1+r) - rN]\mu + \beta(r-1) = 0.$$

Dividing by $\beta(r-1)$:

$$(N-2)\mu^2 - \phi\mu + 1 = 0, \qquad \phi = \frac{rN - \beta(N-1+r)}{\beta(r-1)}.$$

The smaller root is μ_s^{R1} (equation (1)).

A.4 Jump magnitude

At the R1 cutoff, type $\theta = 0$ receives $\beta(1+x)/N$ (aggressive) vs $\beta(r+x)/N$ (conservative) when W proposes. The difference $\beta(r-1)/N$, weighted by $(N-1)/N$ (prob W proposes) and $(1-\mu_s^{R1})$ (prob $\theta = 0$), gives the jump in equation (3).

A.5 Case selection

The R1 screening cutoff lies on the high R2 branch ($\mu_s^{R1} > \mu_s^{R2}$, α-independent) when $\Delta_1(\mu_s^{R2}) > 0$. On the high branch, Δ_1 is a convex quadratic with $\Delta_1(0) > 0$ and $\Delta_1(1) = rN(\beta-1) < 0$, so $\Delta_1(\mu) > 0$ for all $\mu < \mu_s^{R1}$. Hence $\Delta_1(\mu_s^{R2}) > 0$ if and only if $\mu_s^{R2} < \mu_s^{R1}$. Substituting

$\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$ into $\mu_s^{R2} < \mu_s^{R1}$ and solving for α:

$$\frac{\alpha(r-1)}{r-\alpha} < \mu_s^{R1} \quad \Longleftrightarrow \quad \alpha < \frac{\mu_s^{R1} \cdot r}{r-1+\mu_s^{R1}} = \bar{\alpha}(r, \beta, N),$$

which defines the threshold in equation (2).

When $\alpha \geq \bar{\alpha}$, the R1 cutoff falls on the low R2 branch, where $V_W^{R2}(\mu) = (1-\mu)(1-\alpha)/N$ and Δ_1 depends on α through $\omega(\mu)$. The cutoff still exists and is unique: $\Delta_1(0) = \beta(r-1)/N > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$ hold regardless of branch, so a root exists by the intermediate value theorem; since Δ_1 is quadratic on $(0, \mu_s^{R2})$ (at most two roots) with opposite-sign endpoints, there is exactly one root. On the high branch, $\Delta_1 \leq 0$, so no additional roots exist. The jump at μ_s^{R1} remains positive. The proof of Theorem~1 (Appendix~B.5) covers both regimes explicitly: the payoff decomposition $D = D_{\text{base}} + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2} + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}$ is additive regardless of the relative ordering of μ_s^{R1} and μ_s^{R2}, and the endpoint argument establishes $D(\mu) > 0$ on all branches in both cases.

A.6 Budget checks

Under majority, the budget identity $E[V_H] + (N-1)V_W = V_e(\mu)$ holds exactly: $V_H[1 + (N-1)\alpha]/N + (N-1)(1-\alpha)V_e(\mu)/N = V_e(\mu) \cdot [1 + (N-1)\alpha + (N-1)(1-\alpha)]/N = V_e(\mu)$. Under unanimity, the budget identity $E[V_H] + (N-1)V_W = V_e(\mu)$ holds on the conservative R1 branch, where all deals pass in R1. On the aggressive R1 branch, surplus destruction from discounting when $\theta = 1$ rejects in R1 and the game proceeds to R2 yields $E[V_H] + (N-1)V_W = V_e(\mu) - \frac{(N-1)\mu r(1-\beta)}{N}$, so the identity holds with inequality: $E[V_H] + (N-1)V_W \leq V_e(\mu)$.

A.7 Entry thresholds

Under majority, $V_W^{R1}(\mu, M) = \kappa_M V_e(\mu)$ where $\kappa_M = (1-\alpha)[N(N-1) + \beta(q-1)]/(N^2(N-1))$. The entry threshold is $\tau(M) = \max\{0,\ (c/\kappa_M - 1)/(r-1)\}$.

Under unanimity, V_W^{R1} is piecewise. On the conservative R1 branch ($\mu > \mu_s^{R1}$, which also implies $\mu > \mu_s^{R2}$ when $\alpha < \bar{\alpha}$):

$$V_W^{R1}(\mu, U) = \frac{(N+\beta)\, V_e(\mu) - \beta r(1 + N\alpha)}{N^2}.$$

This is affine in μ. Setting $V_W^{R1} = c$ and solving:

$$\tau_U^{con} = \frac{cN^2 - [N + \beta - \beta r(1 + N\alpha)]}{(N + \beta)(r - 1)}. \tag{14}$$

On the aggressive R1 branch ($\mu < \mu_s^{R1}$), the proposer surplus F_1^{agg} is quadratic in μ (the product of $(1 - \mu)$ with the affine terms in $\omega(\mu)$ generates a second-order term). However, the full weak-state expected payoff V_W^{R1}—which averages over proposer and non-proposer roles—simplifies to a piecewise affine expression on each R2 sub-branch, because the quadratic terms in the proposer and non-proposer components cancel (see the proof of Lemma 1 in Appendix B.7). The entry threshold on each sub-branch therefore solves a linear equation.[10]

The numerical results in the paper compute $\tau(U)$ by pointwise evaluation of V_W^{R1}.

Appendix B: Proofs

B.1 Proof of Proposition 1 (Majority produces no screening)

We construct the unique PBE of the two-round Baron-Ferejohn bargaining game under majority rule, proceeding by backward induction from Round 2.

Step 1: Round 2 continuation values. Round 2 is the terminal bargaining round. If no agreement is reached, each player receives its disagreement payoff: H receives $\alpha V(\theta)$; each W_i receives 0.

Case A: H proposes in R2. The hegemon needs $q - 1$ votes from weak states. Each W_i accepts if offered $x_{W_i} \geq 0$ (its disagreement payoff). The cheapest coalition consists of $q - 1$ weak states, each offered $x_{W_i} = 0$. The hegemon keeps the entire pie:

$$V_H^{R2}(\theta, M \mid H \text{ proposes}) = V(\theta).$$

Case B: W_j proposes in R2. The proposer W_j needs $q - 1$ additional votes. Two strategies are

[10] Because V_W^{R1} has a downward jump at μ_s^{R1} (opposite to V_H's upward jump), the formation set $\{\mu : V_W^{R1} \geq c\}$ can be disconnected for intermediate values of c: entry may occur both below μ_s^{R1} (aggressive regime) and above some $\mu_2 > \mu_s^{R1}$ (conservative regime), with a gap around the cutoff. The affine formula (14) gives the entry threshold in the conservative regime and is the relevant quantity when $c > \max_{\mu \leq \mu_s^{R1}} V_W^{R1}(\mu, \text{agg})$.

available: (a) buy $q-1$ other weak states at cost 0 each (their disagreement payoff), total cost 0; or (b) buy H at cost $\alpha V(\theta) > 0$ plus $q-2$ other weak states at cost 0. Since $\alpha V(\theta) > 0$, strategy (a) is strictly cheaper. Hence W_j excludes H from the winning coalition.

Under strategy (a), the proposal passes with q votes. The hegemon receives its disagreement payoff $\alpha V(\theta)$; the proposer keeps $V(\theta) - \alpha V(\theta) = (1-\alpha)V(\theta)$.

R2 continuation values (ex ante over proposer identity). Each player is recognized with probability $1/N$:

$$V_H^{R2}(\theta, M) = \frac{1}{N}V(\theta) + \frac{N-1}{N}\alpha V(\theta) = \frac{V(\theta)[1+(N-1)\alpha]}{N}.$$

$$V_W^{R2}(\mu, M) = \frac{1}{N}(1-\alpha)V_e(\mu) + \frac{N-2}{N}\cdot 0 + \frac{1}{N}\cdot 0 = \frac{(1-\alpha)V_e(\mu)}{N}.$$

Step 2: Round 1 optimal offers.

Case A: H proposes in R1. The hegemon buys $q-1$ weak states at $\beta V_W^{R2}(\mu, M)$ each and keeps the remainder:

$$\Pi_H^{\text{prop}}(\theta, \mu, M) = V(\theta) - (q-1)\beta \cdot \frac{(1-\alpha)V_e(\mu)}{N}.$$

Case B: W_j proposes in R1. The same exclusion logic applies. With $N-2$ other weak states available (excluding W_j and noting $N-2 \geq q-1$ for $N \geq 3$), W_j can form a winning coalition without H. The cost of including H would be at least $\beta V_H^{R2}(\theta = 0, M) = \beta[1+(N-1)\alpha]/N > 0$, which exceeds the cost of one additional weak state $\beta V_W^{R2}(\mu, M) \geq 0$. Hence W_j excludes H, and H receives $\alpha V(\theta)$.

Step 3: Hegemon's expected R1 payoff.

$$\begin{aligned}
E_\theta[V_H^{R1}(\theta, \mu, M)] &= \frac{1}{N}E_\theta\left[V(\theta) - (q-1)\beta\frac{(1-\alpha)V_e(\mu)}{N}\right] + \frac{N-1}{N}\alpha V_e(\mu) && (15)\\
&= \frac{V_e(\mu)}{N}\left[1 - \frac{(q-1)\beta(1-\alpha)}{N}\right] + \frac{(N-1)\alpha V_e(\mu)}{N} && (16)\\
&= \frac{V_e(\mu)}{N^2}\left[N - (q-1)\beta(1-\alpha) + N(N-1)\alpha\right] && (17)\\
&= \lambda_M V_e(\mu), && (18)
\end{aligned}$$

where

$$\lambda_M \equiv \frac{N(1+(N-1)\alpha)-\beta(q-1)(1-\alpha)}{N^2}.$$

Step 4: Affinity in μ. Since $V_e(\mu)=1+\mu(r-1)$ is affine in μ and λ_M is a constant independent of μ, the product $\lambda_M V_e(\mu)$ is affine in μ.

Step 5: No screening. Coalition composition is belief-independent: in both R1 and R2, when W proposes, it forms a majority coalition from other weak states alone, regardless of μ. Because H's vote is never needed, there is no choice between aggressive and conservative offers, no discrete switch in offer structure, and no jump in the hegemon's payoff. The screening mechanism that operates under unanimity is entirely absent.

Step 6: Structural property $\lambda_M > \alpha$.

$$\lambda_M-\alpha=\frac{N(1+(N-1)\alpha)-\beta(q-1)(1-\alpha)-N^2\alpha}{N^2}=\frac{(1-\alpha)[N-\beta(q-1)]}{N^2}>0, \tag{19}$$

since $1-\alpha>0$ (because $\alpha<1/r<1$) and $N-\beta(q-1)>0$ (because $\beta<1$ and $q-1<N$). $\square$

B.2 Proof of Proposition 2 (R1 cutoff existence and uniqueness)

See Appendix A.3 for the derivation. *Existence:* $\Delta_1(0)=\beta(r-1)/N>0$ and $\Delta_1(1)=rN(\beta-1)<0$; by continuity, a root exists in $(0,1)$.

Uniqueness and closed form under $\alpha<\bar{\alpha}$: When $\alpha<\bar{\alpha}(r,\beta,N)$, the root lies on the high R2 branch ($\mu>\mu_s^{R2}$), where Δ_1 is a convex quadratic in μ with coefficients independent of α (Appendix A.3). Since $\Delta_1(\mu_s^{R2})>0$ (Appendix A.5) and $\Delta_1(1)<0$, there is exactly one root in $(\mu_s^{R2},1)$, given by equation (1). Below μ_s^{R2}, Δ_1 is a concave quadratic in μ: the coefficient of μ^2 is $-(N-2)\beta(1-\alpha)/N<0$, arising from the product $(1-\mu)\cdot\omega(\mu)$ in F_1^{agg} where $\omega(\mu)=(N-2)\beta(1-\mu)(1-\alpha)/N$ is linear in μ. Since $\Delta_1(0)=\beta(r-1)/N>0$ and $\Delta_1(\mu_s^{R2})>0$, and a concave function that is positive at both endpoints of an interval is positive throughout, $\Delta_1(\mu)>0$ for all $\mu\in[0,\mu_s^{R2}]$. No additional roots exist.

Uniqueness under $\alpha \geq \bar{\alpha}$: The root falls on the low R2 branch, where Δ_1 is a quadratic in μ that depends on α. On this branch, $\Delta_1(0) > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$ (since $\mu_s^{R2} \geq \mu_s^{R1}$ on the high branch implies $\Delta_1^{\text{high}}(\mu_s^{R2}) \leq 0$, and continuity gives $\Delta_1(\mu_s^{R2}) \leq 0$). Since Δ_1 is quadratic on $(0, \mu_s^{R2})$, it has at most two roots; combined with $\Delta_1(0) > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$, there is exactly one root in $(0, \mu_s^{R2}]$. On the high branch, $\Delta_1 \leq 0$ for $\mu \in [\mu_s^{R2}, 1]$, so no additional roots exist there. □

B.3 Proof of Proposition 3 (Jump)

Direct computation from the difference in H's R1 payoff between the conservative and aggressive branches. See Appendix A.4. □

B.5a Derivation of the payoff decomposition

This appendix derives the decomposition $D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu)$ from the primitive payoff formulas in Appendices A.1–A.3.

Step 0: Primitive payoff components. The first task is to write down the hegemon's R1 payoff from its primitive components. Because proposals are random, the payoff splits naturally by proposer identity, and this separation will prove essential: each component turns out to depend on a different screening regime, which is ultimately why the two corrections are additive.

The hegemon's expected R1 payoff $E[V_H^{R1}(\mu, U)]$ has two components, corresponding to the identity of the R1 proposer.

H proposes (probability $1/N$). The hegemon offers each weak state $\beta V_W^{R2}(\mu)$ (its discounted R2 continuation) and keeps the remainder. The expected payoff across types is:

$$\Pi_H^{\text{prop}}(\mu) = \frac{V_e(\mu) - (N-1)\beta V_W^{R2}(\mu)}{N}. \tag{20}$$

W proposes (probability $(N-1)/N$). The proposing weak state makes an offer to H that depends on the R1 screening regime. Under an aggressive R1 offer ($\mu < \mu_s^{R1}$), only $\theta = 0$ accepts; $\theta = 1$ rejects and the game proceeds to R2 with posterior $\mu' = 1$. The R1 offer to H is $\beta V_H^{R2}(\theta{=}0, \mu'{=}1)$, and $\theta = 1$'s payoff from rejection is $\beta V_H^{R2}(\theta{=}1, \mu'{=}1)$. Since $\mu' = 1 > \mu_s^{R2}$, both R2 values are evaluated in the conservative R2 regime (Appendix A.2, equations (7)–(9)):

$$\beta V_H^{R2}(\theta{=}0, \mu'{=}1) = \frac{\beta(1+x)}{N}, \tag{21}$$

$$\beta V_H^{R2}(\theta{=}1, \mu'{=}1) = \frac{\beta(r+x)}{N}. \tag{22}$$

Under a conservative R1 offer ($\mu > \mu_s^{R1}$), both types accept. The offer is set at $\beta V_H^{R2}(\theta{=}1, \mu'{=}1) = \beta(r+x)/N$ to satisfy the high type's participation constraint. Hence the expected payoff from W's proposal is:

$$\Pi_H^W(\mu) = \frac{N-1}{N} \begin{cases} \mu \dfrac{\beta(r+x)}{N} + (1-\mu)\dfrac{\beta(1+x)}{N} & \text{if } \mu < \mu_s^{R1} \text{ (aggressive R1)}, \\ \dfrac{\beta(r+x)}{N} & \text{if } \mu > \mu_s^{R1} \text{ (conservative R1)}. \end{cases} \tag{23}$$

The total hegemon payoff is $E[V_H^{R1}(\mu, U)] = \Pi_H^{\text{prop}}(\mu) + \Pi_H^W(\mu)$.

Two features are important. First, the R1 offers to H in (21)–(22) are evaluated at the off-path posterior $\mu' = 1$ and therefore do not depend on $V_W^{R2}(\mu)$. Second, H's proposal payoff (20) depends on $V_W^{R2}(\mu)$ but is the same regardless of the R1 regime. These two observations establish that the R1 and R2 corrections are independent.

Step 1: Benchmark payoff (aggressive R1, conservative R2). I now compute the hegemon's total payoff in a benchmark regime—aggressive R1 screening and conservative R2 screening—which serves as the reference point against which the two corrections will be measured. The choice of benchmark is not arbitrary: under this regime, both components of the payoff take their simplest form, and the corrections for each alternative regime can then be isolated one at a time.

Under aggressive R1 and conservative R2 ($V_W^{R2} = [V_e(\mu) - \alpha r]/N$), the two components are:

$$\Pi_H^{\text{prop}}(\mu)\big|_{\text{con-}R2} = \frac{V_e(\mu) - (N-1)\beta[V_e(\mu) - \alpha r]/N}{N} = \frac{N\,V_e(\mu) - (N-1)\beta[V_e(\mu) - \alpha r]}{N^2}, \tag{24}$$

$$\Pi_H^W(\mu)\big|_{\text{agg-}R1} = \frac{(N-1)\beta}{N} \cdot \frac{V_e(\mu) + x}{N} = \frac{(N-1)\beta[V_e(\mu) + x]}{N^2}. \tag{25}$$

Summing:

$$\begin{aligned} E_{\text{bench}}(\mu) &= \frac{N\,V_e(\mu) - (N-1)\beta V_e(\mu) + (N-1)\beta\alpha r + (N-1)\beta V_e(\mu) + (N-1)\beta x}{N^2} \\ &= \frac{N\,V_e(\mu) + (N-1)\beta(\alpha r + x)}{N^2} = \frac{N\,V_e(\mu) + (N-1)\beta N\alpha r}{N^2}, \end{aligned} \tag{26}$$

where the last equality uses $\alpha r + x = \alpha r + (N-1)\alpha r = N\alpha r$.

Step 2: R1 correction δ_{R1}. The next step is to compute the payoff change when the R1 regime switches from aggressive to conservative. Because this switch affects only the R1 offers that W makes to H—and not H's own proposal, which depends on V_W^{R2}—the correction will appear exclusively in Π_H^W, confirming that it can be treated independently of the R2 regime.

When the R1 regime switches from aggressive to conservative ($\mu > \mu_s^{R1}$), Π_H^{prop} is unchanged (it depends only on V_W^{R2}, not on the R1 regime). The change is entirely in Π_H^W. Under aggressive R1:

$$\Pi_H^W\big|_{\text{agg}} = \frac{(N-1)\beta}{N^2}[\mu(r+x) + (1-\mu)(1+x)] = \frac{(N-1)\beta[V_e(\mu)+x]}{N^2}.$$

Under conservative R1:

$$\Pi_H^W\big|_{\text{con}} = \frac{(N-1)\beta(r+x)}{N^2}.$$

The correction is:

$$\delta_{R1}(\mu) \equiv \Pi_H^W\big|_{\text{con}} - \Pi_H^W\big|_{\text{agg}} = \frac{(N-1)\beta}{N^2}[(r+x) - V_e(\mu) - x] = \frac{(N-1)\beta(r-1)(1-\mu)}{N^2}. \tag{27}$$

This is non-negative for all $\mu \leq 1$, with equality only at $\mu = 1$. The correction reflects the overpayment that the low type receives under conservative R1 offers.

Step 3: R2 correction δ_{R2}. I now derive the payoff change when the R2 regime switches from conservative to aggressive. By an argument symmetric to Step 2, this switch affects only H's proposal payoff Π_H^{prop}—the R1 offers to H are pinned at the off-path posterior $\mu' = 1$, which always lies in the conservative R2 regime regardless of the on-path belief μ.

When the R2 regime switches from conservative to aggressive ($\mu < \mu_s^{R2}$), Π_H^W is unchanged (the R1 offers to H are evaluated at $\mu' = 1$, which is always in the conservative R2 regime). The change is entirely in Π_H^{prop}. Substituting $V_W^{R2,\text{agg}}(\mu) = (1-\mu)(1-\alpha)/N$ (Appendix A.2) in place of $V_W^{R2,\text{con}}(\mu) = [V_e(\mu) - \alpha r]/N$:

$$\begin{aligned}\delta_{R2}(\mu) &\equiv \Pi_H^{\text{prop}}\big|_{\text{agg-}R2} - \Pi_H^{\text{prop}}\big|_{\text{con-}R2} \\ &= \frac{-(N-1)\beta}{N^2}\Big[(1-\mu)(1-\alpha) - V_e(\mu) + \alpha r\Big] \\ &= \frac{(N-1)\beta}{N^2}\Big[V_e(\mu) - \alpha r - (1-\mu)(1-\alpha)\Big]. \end{aligned} \tag{28}$$

Expanding the bracket:

$$V_e(\mu) - \alpha r - (1-\mu)(1-\alpha) = \mu(r-\alpha) - \alpha(r-1),$$

so $\delta_{R2}(\mu) = (N-1)\beta[\mu(r-\alpha) - \alpha(r-1)]/N^2$, which matches the formula in Appendix B.5. At $\mu = \mu_s^{R2} = \alpha(r-1)/(r-\alpha)$, direct substitution gives $\delta_{R2}(\mu_s^{R2}) = 0$, confirming continuity at the R2 cutoff.

Step 4: Additivity. With both corrections in hand, I can now establish that they combine additively. This is the key structural result: it means the full payoff on any branch of the belief space can be written as the benchmark plus whichever corrections are active, without interaction terms.

The two corrections are independent because they affect disjoint components of H's payoff:

- δ_{R2} operates through Π_H^{prop} (equation (20)), which depends on $V_W^{R2}(\mu)$ but is invariant to the R1 regime.
- δ_{R1} operates through Π_H^W (equation (23)), which depends on the R1 offer type but is invariant to $V_W^{R2}(\mu)$—because the R1 offers to H are pinned at $\beta V_H^{R2}(\theta, \mu'{=}1)$, which is always evaluated at $\mu' = 1 > \mu_s^{R2}$.

Hence the full unanimity payoff is:

$$E[V_H^{R1}(\mu, U)] = E_{\text{bench}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu).$$

Step 5: Subtracting majority and collecting terms. The final step subtracts the majority payoff from the unanimity benchmark to obtain D_{base}. This completes the decomposition: the payoff difference $D(\mu)$ on any branch equals D_{base} plus the active corrections, and all three components are affine in μ.

Under majority (Appendix A.1), $E[V_H^{R1}(\mu, M)] = \lambda_M V_e(\mu)$ where $\lambda_M = [N(1+(N-1)\alpha) - C_{\text{buy}}]/N^2$ with $C_{\text{buy}} = \beta(q-1)(1-\alpha)$ and $q = \lfloor N/2 \rfloor + 1$. Define $C_{\text{out}} = N(N-1)\alpha$. Then:

$$\begin{aligned} D_{\text{base}}(\mu) &\equiv E_{\text{bench}}(\mu) - \lambda_M V_e(\mu) \\ &= \frac{N\,V_e(\mu) + (N-1)\beta N \alpha r}{N^2} - \frac{N(1+(N-1)\alpha) - C_{\text{buy}}}{N^2} V_e(\mu) \\ &= \frac{V_e(\mu)[N - N - C_{\text{out}} + C_{\text{buy}}] + C_{\text{out}}\beta r}{N^2} \\ &= \frac{(C_{\text{buy}} - C_{\text{out}})V_e(\mu) + C_{\text{out}}\beta r}{N^2}, \end{aligned} \tag{29}$$

where the third line uses $N(1+(N-1)\alpha) = N + C_{\text{out}}$ and $(N-1)\beta N \alpha r = C_{\text{out}}\beta r$.

The payoff difference is therefore:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu),$$

with D_{base}, δ_{R2}, δ_{R1} as in equations (29), (28), (27). □

B.5 Proof of Theorem 1 (Conditional payoff dominance)

The payoff difference $D(\mu)$ decomposes into a baseline affine term plus two correction terms, each itself affine. Positivity therefore reduces to checking endpoints. The decomposition is derived in Appendix B.5a.

Let $C_{\text{buy}} \equiv \beta(q-1)(1-\alpha)$ (the vote-buying cost under majority) and $C_{\text{out}} \equiv N(N-1)\alpha$ (the aggregate outside-option cost). Under majority, $E[V_H^{R1}(\mu, M)] = \lambda_M V_e(\mu)$ where $\lambda_M = [N(1+(N-1)\alpha) - C_{\text{buy}}]/N^2$.

Step 1: Decomposition. The decomposition has three components, each affine in μ. Because an affine function that is positive at both endpoints of an interval is positive throughout, it suffices to check a finite number of boundary values.

Define:

$$D_{\text{base}}(\mu) \equiv \frac{(C_{\text{buy}} - C_{\text{out}})V_e(\mu) + C_{\text{out}}\beta r}{N^2}, \tag{30}$$

$$\delta_{R2}(\mu) \equiv \frac{(N-1)\beta[\mu(r-\alpha) - \alpha(r-1)]}{N^2}, \tag{31}$$

$$\delta_{R1}(\mu) \equiv \frac{(N-1)\beta(r-1)(1-\mu)}{N^2}. \tag{32}$$

The difference $D(\mu) \equiv E[V_H^{R1}(\mu, U)] - \lambda_M V_e(\mu)$ decomposes as:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu).$$

This decomposition holds regardless of the relative ordering of μ_s^{R1} and μ_s^{R2}. The two corrections are additive because they affect independent components of H's payoff: δ_{R2} captures the change in $V_W^{R2}(\mu)$ (which enters through H's proposal offer), while δ_{R1} captures the change in the R1 offer H receives when W proposes (which depends on the R1 regime but not on V_W^{R2}).

D_{base} is the difference under aggressive R1 and conservative R2. The R2 correction vanishes at the screening cutoff: $\delta_{R2}(\mu_s^{R2}) = 0$, ensuring continuity at μ_s^{R2}. The R1 correction satisfies $\delta_{R1}(\mu) \geq 0$ for all $\mu \leq 1$, with equality only at $\mu = 1$.

The proof strategy is as follows. D_{base} is positive everywhere on $[0, 1]$ (Step 2). The R2 correction δ_{R2} is active only below μ_s^{R2}; it is negative at $\mu = 0$ but vanishes at μ_s^{R2}, so the tighter endpoint condition is $D_I(0) = D_{\text{base}}(0) + \delta_{R2}(0) > 0$. The R1 correction δ_{R1} is active only above μ_s^{R1} and is non-negative throughout, so it can only help. Step 3 assembles these facts branch by branch.

Step 2: $D_{\text{base}} > 0$ on $[0, 1]$. Since D_{base} is affine in μ—it depends on μ only through $V_e(\mu) = 1 + \mu(r-1)$, which is itself affine—checking positivity at the two endpoints $\mu = 0$ and $\mu = 1$ is both necessary and sufficient.

At $\mu = 1$:

$$D_{\text{base}}(1) = \frac{r[C_{\text{buy}} - C_{\text{out}}(1-\beta)]}{N^2}.$$

Setting $C_{\text{buy}} - C_{\text{out}}(1-\beta) = 0$ and solving for α yields $\alpha = \alpha^*(N, \beta)$. Since $\alpha < \alpha^*$, $D_{\text{base}}(1) > 0$.

At $\mu = 0$: Define $D_I(\mu) \equiv D_{\text{base}}(\mu) + \delta_{R2}(\mu)$, the payoff difference on the low-belief branch. We check $D_I(0)$ rather than $D_{\text{base}}(0)$ directly, because $\delta_{R2}(0) < 0$—so $D_I(0)$ is the tighter endpoint condition. Write $D_I(0) = [\beta(q-1) - \alpha d_0]/N^2$ where $d_0 = N(N-1) - \beta((N-1)^2 r + N - q)$. If $d_0 \leq 0$, then $D_I(0) \geq \beta(q-1)/N^2 > 0$. If $d_0 > 0$, the threshold $\bar{\alpha}_0 = \beta(q-1)/d_0$ satisfies $\bar{\alpha}_0 > \alpha^*$ because $d_0 < d_* \equiv N(N-1) - \beta(N^2 - N - q + 1)$ (since $d_* - d_0 = \beta(N-1)^2(r-1) > 0$). Hence $\alpha < \alpha^* < \bar{\alpha}_0$ implies $D_I(0) > 0$, and therefore $D_{\text{base}}(0) > D_I(0) > 0$.

Conclusion: D_{base} is affine with positive values at both endpoints, hence $D_{\text{base}}(\mu) > 0$ for all $\mu \in [0, 1]$.

Step 3: Assembly. The strategy is to partition the belief space $(0, 1]$ by the two screening cutoffs and verify $D > 0$ on each branch, using the endpoint results from Steps 1–2. On every branch, D is a sum of affine functions and is therefore itself affine, so positivity at branch endpoints implies positivity throughout.

The standard case is $\mu_s^{R2} \leq \mu_s^{R1}$, which partitions $(0, 1]$ into three branches.

Low-belief branch ($\mu < \mu_s^{R2}$): $D = D_I(\mu)$, which is affine. At the left endpoint, $D_I(0) > 0$ (Step 2). At the right endpoint, $\delta_{R2}(\mu_s^{R2}) = 0$, so $D(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) > 0$ (Step 2). Affine with positive endpoints implies $D > 0$ throughout.

Middle branch ($\mu_s^{R2} \leq \mu \leq \mu_s^{R1}$): $D = D_{\text{base}} > 0$ directly from Step 2.

High-belief branch ($\mu > \mu_s^{R1}$): $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$, since $\delta_{R1} \geq 0$.

Alternative case ($\mu_s^{R1} < \mu_s^{R2}$). The additive decomposition remains valid because the two corrections affect independent components of H's payoff (Step 1). The partition of $(0, 1]$ now has three branches:

Low-belief branch ($\mu < \mu_s^{R1}$): aggressive R1, aggressive R2. $D = D_I(\mu) = D_{\text{base}}(\mu) + \delta_{R2}(\mu)$, which is affine. At $\mu = 0$, $D_I(0) > 0$ (Step 2). At $\mu = \mu_s^{R1}$, D_I is continuous from the left and $D_I(\mu_s^{R1}) > 0$ because D_I is affine with $D_I(0) > 0$ and $D_I(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) > 0$ (Step 2), and $\mu_s^{R1} < \mu_s^{R2}$.

Middle branch ($\mu_s^{R1} \leq \mu < \mu_s^{R2}$): conservative R1, aggressive R2. Both corrections are active: $D = D_{\text{base}} + \delta_{R2} + \delta_{R1}$. This is affine. At μ_s^{R1}, D jumps up by $\delta_{R1}(\mu_s^{R1}) > 0$ relative to the low branch, so $D(\mu_s^{R1}) = D_I(\mu_s^{R1}) + \delta_{R1}(\mu_s^{R1}) > D_I(\mu_s^{R1}) > 0$. At μ_s^{R2}, $\delta_{R2}(\mu_s^{R2}) = 0$, so $D(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) + \delta_{R1}(\mu_s^{R2}) \geq D_{\text{base}}(\mu_s^{R2}) > 0$. Affine with positive values at both endpoints implies $D > 0$ throughout.

High-belief branch ($\mu \geq \mu_s^{R2}$): conservative R1, conservative R2. $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$, since $\delta_{R1} \geq 0$.

Combining all branches in either case: $D(\mu) > 0$ for every $\mu \in (0, 1]$. This establishes sufficiency.

Step 4: Necessity. To establish that $\alpha < \alpha^*$ is not merely sufficient but also necessary, I show that the inequality $D(\mu) > 0$ fails at $\mu = 1$ whenever $\alpha \geq \alpha^*$.

At $\mu = 1$, both corrections vanish ($\delta_{R1}(1) = 0$ and $\mu = 1 > \mu_s^{R2}$), so $D(1) = D_{\text{base}}(1) = r[C_{\text{buy}} - C_{\text{out}}(1 - \beta)]/N^2$. Since α^* is the root of $C_{\text{buy}} - C_{\text{out}}(1 - \beta) = 0$, $\alpha \geq \alpha^*$ implies $D(1) \leq 0$, and the strict inequality fails. $\square$

B.6 Proof of Corollary (Unanimity dominance on the formation set)

Fix $p \in \mathcal{F}_U$, so $V_W^{R1}(p, U) \geq c$. We first show that $p \in \mathcal{F}_M$. Under majority, the budget identity holds exactly: $E[V_H(\mu, M)] + (N-1)V_W(\mu, M) = V_e(\mu)$. Under unanimity, surplus destruction from discounting on the aggressive branch yields $E[V_H(\mu, U)] + (N - 1)V_W(\mu, U) \leq V_e(\mu)$. Since Theorem 1 gives $E[V_H(\mu, U)] > E[V_H(\mu, M)]$, it follows that $V_W(\mu, U) < V_W(\mu, M)$ for every $\mu \in (0, 1]$. Hence $V_W^{R1}(p, M) > V_W^{R1}(p, U) \geq c$, so entry occurs under majority at p. In particular, $\mathcal{F}_U \subseteq \mathcal{F}_M$.

Since the institution forms under both rules at p, the hegemon's payoff under each rule is the expected bargaining payoff: $V_H(R, p) = E_\theta[V_H^{R1}(\theta, p, R)]$. By Theorem 1, $V_H(U, p) > V_H(M, p)$. $\square$

B.7 Lemma: Global maximum of V_W^{R1} under unanimity

Lemma 1 (Global maximum of weak-state payoff)**.** $V_W^{R1}(\mu, U) \leq V_W^{R1}(1, U) = r(1 - \beta\alpha)/N$ *for every* $\mu \in (0, 1]$*, with equality only at* $\mu = 1$*. Hence if* $\mathcal{F}_U \neq \emptyset$*, then* $1 \in \mathcal{F}_U$*.*

Proof. At $\mu = 1$, the equilibrium R1 offer is conservative: $F_1^{con}(1) = r - \beta(r + x)/N - \omega(1) > 0$

while $F_1^{agg}(1) = \beta r(1-\alpha)/N > 0$, and the conservative surplus exceeds the aggressive surplus because $\beta < 1$ implies the full pie exceeds the discounted R2 value. Hence $\bar{V}_W \equiv V_W^{R1}(1, U) = r(1-\beta\alpha)/N$.

We bound every feasible R1 payoff candidate by $\bar{V}_W$, using the full ex ante weak-state payoff (not only the proposer surplus). Let $W_2^L(\mu) = (1-\mu)(1-\alpha)/N$ and $W_2^H(\mu) = [V_e(\mu) - \alpha r]/N$ denote the R2 continuation values on the low and high branches, $W_2^1 = r(1-\alpha)/N$ the R2 value at $\mu = 1$, and $x = (N-1)\alpha r$. Under a conservative R1 offer (all types accept), a non-proposing weak state receives $\beta W_2^j(\mu)$ whenever another player proposes, giving $V_W^{Cj}(\mu) = F_1^{con,j}(\mu)/N + (N-1)\beta W_2^j(\mu)/N$. Under an aggressive R1 offer ($\theta = 1$ rejects, game proceeds to R2 with posterior one), $V_W^{Aj}(\mu) = F_1^{agg,j}(\mu)/N + \beta W_2^j(\mu)/N + (N-2)[(1-\mu)\beta W_2^j(\mu) + \mu\beta W_2^1]/N$. The equilibrium payoff $V_W^{R1}(\mu, U)$ is one of the four candidates V_W^{CH}, V_W^{AH}, V_W^{CL}, V_W^{AL}, depending on the R1 offer type and the R2 branch.

Conservative R1, high R2 ($\mu \geq \mu_s^{R2}$): $V_W^{CH}(\mu) = [(N+\beta)V_e(\mu) - \beta r(1+N\alpha)]/N^2$, so $\bar{V}_W - V_W^{CH}(\mu) = (N+\beta)(r-1)(1-\mu)/N^2 \geq 0$, with equality only at $\mu = 1$.

Aggressive R1, high R2 ($\mu \geq \mu_s^{R2}$): The proposer surplus $F_1^{agg,H}$ contains a term $(1-\mu)\beta W_2^H(\mu)$ that is quadratic in μ, but this term cancels exactly with the corresponding non-proposer term $(1-\mu)\beta W_2^H(\mu)$ in V_W^{AH}, leaving a result that is affine. After substituting $W_2^H(\mu) = [V_e(\mu) - \alpha r]/N$ and simplifying: $\bar{V}_W - V_W^{AH}(\mu) = [(1-\mu)(r-1) + \mu r(1-\beta)]/N > 0$ for all $\mu \in [0, 1]$.

Conservative R1, low R2 ($\mu < \mu_s^{R2}$): Using $W_2^L(\mu) = (1-\mu)(1-\alpha)/N$, $\bar{V}_W - V_W^{CL}(\mu)$ is affine in μ. At $\mu = 0$: $(r-1)(N - \alpha\beta + \beta)/N^2 > 0$. At $\mu = \mu_s^{R2}$: $r(r-1)(1-\alpha)(N+\beta)/[N^2(r-\alpha)] > 0$. Affine with positive endpoints implies $\bar{V}_W - V_W^{CL} > 0$ on $[0, \mu_s^{R2}]$.

Aggressive R1, low R2 ($\mu < \mu_s^{R2}$): The same quadratic cancellation as in the high-R2 case applies: the $(1-\mu)\beta W_2^L(\mu)$ terms in the proposer and non-proposer components cancel, and $\bar{V}_W - V_W^{AL}(\mu)$ is affine. At $\mu = 0$: $(r-1)(N-\alpha\beta)/N^2 > 0$. At $\mu = \mu_s^{R2}$: $r(r-1)(1-\alpha\beta)/[N(r-\alpha)] > 0$. Hence $\bar{V}_W - V_W^{AL} > 0$ on $[0, \mu_s^{R2}]$.

Since every payoff candidate is strictly below $\bar{V}_W$ for $\mu < 1$, and the equilibrium payoff selects one of these candidates, $V_W^{R1}(\mu, U) < \bar{V}_W$ for all $\mu < 1$. The corollary follows: if $V_W^{R1}(\mu', U) \geq c$

for some μ', then $V_W^{R1}(1, U) \geq V_W^{R1}(\mu', U) \geq c$, so $1 \in \mathcal{F}_U$. □ □

B.8 Proof of Proposition 4 (Institutional classification)

We verify each case exhaustively. Since $\mathcal{F}_U \subseteq \mathcal{F}_M$ (Corollary, Appendix B.6), the three cases $p \in \mathcal{F}_U$, $p \in \mathcal{F}_M \setminus \mathcal{F}_U$, and $p \notin \mathcal{F}_M$ partition $(0, 1]$.

Case (i): $p \in \mathcal{F}_U$**.** The institution forms under both rules. By Corollary~1, $V_H(U, p) > V_H(M, p)$, so $U \succ M$.

Case (ii): $p \in \mathcal{F}_M \setminus \mathcal{F}_U$**.** The institution forms under majority but not under unanimity. Under unanimity, since $p \notin \mathcal{F}_U$, the institution does not form and the hegemon receives its bilateral outside option: $V_H(U, p) = \alpha V_e(p)$. Under majority, since $p \in \mathcal{F}_M$, the institution forms and, by Proposition 1, $V_H(M, p) = \lambda_M V_e(p)$. Since $\lambda_M > \alpha$ (Appendix B.1, Step 6) and $V_e(p) > 0$, we have $V_H(M, p) = \lambda_M V_e(p) > \alpha V_e(p) = V_H(U, p)$. Hence $M \succ U$.

Case (iii): $p \notin \mathcal{F}_M$**.** The institution does not form under either rule (since $\mathcal{F}_U \subseteq \mathcal{F}_M$, $p \notin \mathcal{F}_M$ implies $p \notin \mathcal{F}_U$). Both rules yield the outside-option payoff $\alpha V_e(p)$, so $U \sim M$. □

Appendix C: Continuous Types

This appendix extends the model to continuous types. Section C.1 establishes that the screening rent is strictly positive for any continuous distribution with full support—a distribution-free result. Sections C.2–C.4 derive closed-form payoff expressions for the uniform distribution and show that the dominance threshold α^*_{cont} weakly exceeds the binary-type threshold α^*.

C.1 Screening rent for general distributions

The model is identical to the main model except that θ is drawn from a continuous distribution F on $[1, r]$, $r > 1$, with full support and positive density f on $(1, r)$. The cooperation value is $V(\theta) = \theta$, and the outside-option constraint becomes $\alpha \in (0, 1/r)$.

Proposition 5 (Screening rent under continuous types)**.** *(a) In R2, W's optimal screening cutoff* *satisfies* $\theta_2^* > 1$*, and H's screening rent is*

$$Rent_{R2} = \int_1^{\theta_2^*} \alpha(\theta_2^* - \theta)\, dF(\theta) > 0.$$

(b) In R1, H's screening rent from W's proposal is strictly positive. (c) Under majority, W's proposals generate zero screening rent in both rounds.

Proof. **R2 under unanimity.** When W proposes in R2, W sets $y_H = \alpha\theta^*$ for some cutoff $\theta^* \in [1, r]$; types $\theta \leq \theta^*$ accept and types $\theta > \theta^*$ reject. W's expected payoff is $\pi_W(\theta^*) = \int_1^{\theta^*} (\theta - \alpha\theta^*)\, dF(\theta)$.

Optimal cutoff $\theta_2^* > 1$. Since $\pi_W(1) = 0$ and, for small $\epsilon > 0$, $\pi_W(1+\epsilon) \geq (1-\alpha(1+\epsilon))F(1+\epsilon) > 0$ (using $\alpha < 1/r < 1$ and full support), the global maximizer cannot be at $\theta^* = 1$.

Screening rent. H's expected payoff from W's R2 proposal is $\int_1^{\theta_2^*} \alpha\theta_2^*\, dF(\theta) + \int_{\theta_2^*}^{r} \alpha\theta\, dF(\theta)$. The rent relative to the perfect-discrimination benchmark $\int_1^{r} \alpha\theta\, dF(\theta)$ is

$$\text{Rent}_{R2} = \int_1^{\theta_2^*} \alpha(\theta_2^* - \theta)\, dF(\theta) > 0.$$

Positivity: the integrand $\alpha(\theta_2^* - \theta) > 0$ for $\theta \in [1, \theta_2^*)$, and F assigns positive mass to $[1, \theta_2^*)$ since $\theta_2^* > 1$ and F has full support.

R1 under unanimity. $V_H^{R2}(\theta, U)$ is strictly increasing in θ: the H-proposes component (θ/N) is strictly increasing, and the W-proposes component is non-decreasing. In R1, W offers y_H and type θ accepts iff $y_H \geq \beta V_H^{R2}(\theta, U)$. Strict monotonicity yields a cutoff θ_1^*, and the same argument as R2 gives $\theta_1^* > 1$ and

$$\text{Rent}_{R1} = \int_1^{\theta_1^*} [\beta V_H^{R2}(\theta_1^*, U) - \beta V_H^{R2}(\theta, U)] dF(\theta) > 0.$$

Majority. Under majority, W excludes H from the winning coalition. H receives $\alpha\theta$ type by type—the perfect-discrimination benchmark—so the screening rent is zero in both rounds. $\square$

C.2 Uniform distribution: R2 equilibrium

For the remainder of this appendix, let $F = \text{Uniform}[1, r]$, so $f(\theta) = 1/(r-1)$, $F(\theta) = (\theta - 1)/(r-1)$, and $E[\theta] = (1+r)/2$. Define $x \equiv (N-1)\alpha r$.

R2 under unanimity: $\theta_2^* = r$. W's profit derivative is $\pi_W'(\theta^*) = [(1-2\alpha)\theta^* + \alpha]/(r-1)$. The root $\theta^* = \alpha/(2\alpha-1)$ satisfies $\theta^* \leq r$ only if $(r-1)^2 \leq 0$, which is impossible for $r > 1$. Hence $\pi_W' > 0$ on $[1, r]$ and $\theta_2^* = r$: W pools all types with the offer $y_H = \alpha r$. The same argument applies to any truncation $\text{Uniform}[a, r]$ with $a \geq 1$, so $\theta_2^* = r$ holds regardless of the R1 screening cutoff.

R2 continuation values:

$$V_H^{R2}(\theta, U) = \frac{\theta + x}{N}, \qquad V_W^{R2}(\theta, U) = \frac{\theta - \alpha r}{N}, \tag{33}$$

$$V_H^{R2}(\theta, M) = \frac{\theta(1 + (N-1)\alpha)}{N}, \qquad V_W^{R2}(\theta, M) = \frac{(1-\alpha)\theta}{N}. \tag{34}$$

C.3 R1 equilibrium

H **proposes under unanimity** (probability $1/N$). H offers each W its discounted R2 continuation value $\beta E_\theta[V_W^{R2}(\theta, U)] = \beta(1 + r - 2\alpha r)/(2N)$ and keeps

$$V_H^{R1}(\theta,\ H\text{-prop},\ U) = \theta - \frac{(N-1)\beta(1 + r - 2\alpha r)}{2N}. \tag{35}$$

W **proposes under unanimity** (probability $(N-1)/N$). The proposer W_k chooses a screening cutoff θ_1^*: types $\theta \leq \theta_1^*$ accept the offer $y_H = \beta(\theta_1^* + x)/N$; types $\theta > \theta_1^*$ reject, and R2 begins with posterior $\text{Uniform}[\theta_1^*, r]$ (under which $\theta_2^* = r$ still holds, as noted above).

Under unanimity, W_k must also secure votes from $N-2$ non-proposing weak states. Setting each W_i's expected payoff from accepting equal to its rejection payoff (the discounted R2 continuation under the prior) and solving gives the binding offer

$$y_{W_i}(\theta_1^*) = \frac{\beta(\theta_1^* + 1 - 2\alpha r)}{2N}. \tag{36}$$

Substituting into W_k's expected payoff from cutoff $s \equiv \theta_1^*$:

$$\Pi_{W_k}(s) = \frac{1}{r-1}\int_1^s \left[\theta - \frac{\beta(Ns + N - 2 + 2\alpha r)}{2N}\right] d\theta + \frac{r-s}{r-1} \cdot \frac{\beta[s + (1-2\alpha)r]}{2N}, \quad (37)$$

where the first term is the proposer's surplus when H accepts (types $\theta \leq s$) and the second is the discounted R2 payoff when H rejects (types $\theta > s$). Multiplying by $2N(r-1)$ and collecting:

$$2N(r-1)\Pi_{W_k}(s) = \underbrace{[N - (N+1)\beta]}_{s^2 \text{ coeff.}}\, s^2 + \underbrace{2\beta}_{s \text{ coeff.}}\, s + (\text{terms in } \alpha \text{ only}).$$

Since α enters only the constant term, the optimal cutoff is independent of α:

$$\theta_1^* = \begin{cases} r & \text{if } \beta \leq \dfrac{Nr}{(N+1)r-1}, \\ \min\left(r,\ \dfrac{\beta}{(N+1)\beta - N}\right) & \text{otherwise.} \end{cases} \quad (38)$$

When $\beta \leq Nr/[(N+1)r-1]$, the unconstrained maximum of the quadratic is at $s^* \geq r$, so $\theta_1^* = r$ (full pooling). When $\beta > Nr/[(N+1)r-1]$, the interior maximum $s^* = \beta/[(N+1)\beta - N] < r$ binds.

H**'s payoff from** W**'s proposal.** Given cutoff θ_1^*, types $\theta \leq \theta_1^*$ receive $\beta(\theta_1^* + x)/N$ and types $\theta > \theta_1^*$ receive $\beta(\theta + x)/N$. Integrating over Uniform$[1, r]$:

$$\begin{aligned} E_\theta[V_H^{R1} \mid W\text{-prop},\ U] &= \frac{\beta}{N(r-1)}\left[(\theta_1^* - 1)(\theta_1^* + x) + \frac{r^2 - (\theta_1^*)^2}{2} + (r - \theta_1^*)x\right] \\ &= \frac{\beta}{N}\left[\frac{(\theta_1^* - 1)^2}{2(r-1)} + \frac{1+r}{2} + x\right]. \end{aligned} \quad (39)$$

The term $(\theta_1^* - 1)^2/[2(r-1)]$ is the screening rent from R1 pooling. It is zero when $\theta_1^* = 1$ and maximal at $\theta_1^* = r$, giving $(r-1)/2$. This term is absent under majority. Verification: at $\theta_1^* = r$, equation (39) gives $\beta(r+x)/N = \beta V_H^{R2}(r, U)$.

H **proposes under majority** (probability $1/N$). H buys $q-1$ votes at $\beta E_\theta[V_W^{R2}(\theta, M)] =$

$\beta(1-\alpha)(1+r)/(2N)$:

$$E_\theta[V_H^{R1}(\theta,\ H\text{-prop},\ M)] = \frac{1+r}{2}\left[1 - \frac{(q-1)\beta(1-\alpha)}{N}\right]. \tag{40}$$

W **proposes under majority** (probability $(N-1)/N$). W excludes H from the winning coalition; H captures $\alpha\theta$. Expected payoff: $\alpha(1+r)/2$.

C.4 Payoff comparison and dominance threshold

Combined R1 payoff under unanimity. Weighting H-proposes by $1/N$ and W-proposes by $(N-1)/N$, and using $\alpha r + x = N\alpha r$:

$$E_\theta[V_H^{R1}(U)] = \frac{1+r}{2N} + \frac{(N-1)\beta}{N^2}\left[\frac{(\theta_1^*-1)^2}{2(r-1)} + N\alpha r\right]. \tag{41}$$

Combined R1 payoff under majority. With $C_{\text{buy}} \equiv \beta(q-1)(1-\alpha)$ and $C_{\text{out}} \equiv N(N-1)\alpha$:

$$E_\theta[V_H^{R1}(M)] = \frac{(1+r)(N + C_{\text{out}} - C_{\text{buy}})}{2N^2} = \lambda_M \cdot \frac{1+r}{2}, \tag{42}$$

where $\lambda_M = (N + C_{\text{out}} - C_{\text{buy}})/N^2$, as in the binary model (Appendix B.1).

Payoff difference. The difference $D \equiv E_\theta[V_H^{R1}(U)] - E_\theta[V_H^{R1}(M)]$ is

$$D = \frac{1}{2N^2}\left[\frac{(N-1)\beta(\theta_1^*-1)^2}{r-1} + (1+r)C_{\text{buy}} - C_{\text{out}}(1+r-2\beta r)\right]. \tag{43}$$

D is affine in α. Expanding C_{buy} and C_{out}:

$$2N^2 D = \underbrace{\frac{(N-1)\beta(\theta_1^*-1)^2}{r-1} + (1+r)\beta(q-1)}_{\equiv A > 0} - \alpha\,\underbrace{\left[(1+r)\beta(q-1) + N(N-1)(1+r-2\beta r)\right]}_{\equiv \Lambda}.$$

The threshold $D = 0$ gives the dominance boundary:

$$\alpha^*_{\text{cont}}(N, \beta, r) = \frac{A}{\Lambda} = \frac{\dfrac{(N-1)\beta(\theta_1^* - 1)^2}{r-1} + (1+r)\beta(q-1)}{(1+r)\beta(q-1) + N(N-1)(1+r-2\beta r)}. \tag{44}$$

When $\Lambda > 0$, unanimity dominates ($D > 0$) for $\alpha < \alpha^*_{\text{cont}}$. When $\Lambda \leq 0$, unanimity dominates for all $\alpha \in (0, 1/r)$.

Structure of D. The three components have transparent interpretations. The screening-rent term $(N-1)\beta(\theta_1^* - 1)^2/(r-1) \geq 0$ captures the overpayment to low types when W pools in R1 under unanimity; it is absent under majority. The vote-buying term $(1+r)C_{\text{buy}} > 0$ reflects H's cheaper coalition formation under majority ($q-1$ votes instead of $N-1$). The outside-option term $-C_{\text{out}}(1+r-2\beta r)$ captures the net effect of H's bilateral alternative: under majority, H captures $\alpha\theta$ when excluded; under unanimity, H captures a pooling offer. The sign depends on β and r.

Comparison with the binary-type threshold. The binary-type threshold (Theorem 1) is

$$\alpha^* = \frac{\beta(q-1)}{N(N-1)(1-\beta) + \beta(q-1)}.$$

Limit $r \to 1^+$. As $r \to 1$, the type space degenerates: $\theta_1^* \to 1$, the screening rent vanishes, and direct computation gives

$$\alpha^*_{\text{cont}} \to \frac{\beta(q-1)}{\beta(q-1) + N(N-1)(1-\beta)} = \alpha^*.$$

The two thresholds coincide: the continuous model recovers the binary result when the type space becomes trivial.

For $r > 1$. The numerator A of α^*_{cont} exceeds the (scaled) numerator of α^* by the screening-rent term $(N-1)\beta(\theta_1^* - 1)^2/(r-1) > 0$. A grid search over $N \in \{3, 5, 7, 10, 20, 50, 100, 164\}$, $\beta \in \{0.10, 0.15, \ldots, 0.99\}$, $r \in \{1.01, 1.1, 1.5, 2, 3, 5, 10\}$ (1,008 combinations) confirms

$$\alpha^*_{\text{cont}} \geq \alpha^* \quad \text{for all parameter values,}$$

with equality only at $r = 1$. The minimum ratio is $\alpha^*_{\text{cont}}/\alpha^* = 1.010$ (at $r = 1.01$). Unconditional dominance ($\Lambda \leq 0$, implying $D > 0$ for all $\alpha \in (0, 1/r)$) occurs in 32% of the grid. Analytical formulas are verified against Monte Carlo simulation (10^6 draws per parameter combination) to relative error below 5×10^{-4}.

The binary model is therefore the most conservative specification for the mechanism: continuous types amplify the screening rent, enlarging the parameter region where unanimity dominates majority.

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